

Lesson 10

Apply Analytical Thinking & Problem Solving into Data Analysis Cases

Turn messy problems into structured, solvable analytics challenges

March 2025

Why Analytical Thinking Matters in Data Analysis?

Structure your thinking before crunching your numbers.

What Do We Mean by Analytical Thinking?

Analytical thinking is the ability to break down complex problems, evaluate them from multiple angles, and use data and reasoning to make the best possible decision.

▷ **Data-driven decision-making (DDDM) is built on a strong foundation of three core elements:**



Data

(The raw material)

- Comes from various sources (databases, APIs, surveys, IoT, etc.).
 - Needs to be cleaned, processed, and structured before analysis.
- Example: Sales transaction records, customer feedback, website clicks.*



Statistics

(Extracting Meaning from Data)

- Helps summarize, interpret, and find patterns in data.
 - Enables decision-making through probabilities, trends, and correlations.
- Example: Using regression analysis to predict next month's sales.*



Domain Knowledge

(Context Matters)

- Understanding industry-specific trends ensures correct interpretation
- Example: In finance, a sudden drop in stock prices may not be alarming if it's due to seasonal trends.*

Analytical Thinking & Problem Solving

(The ability to frame problems, ask the right questions, and derive insights.)

▷ **Data-driven decision-making (DDDM) is built on a strong foundation of three core elements:**



Data

(The raw material)

provides the foundation



Statistics

(Extracting Meaning from Data)

ensures accuracy & trends.



Domain Knowledge

(Context Matters)

makes insights relevant.

turns insights into actions.

Analytical Thinking.

Break Down → Analyze → Solve.



Synthesis:

Gathering and connecting relevant information



Systemization:

Organizing facts into a logical structure



Critical Thinking

Challenging assumptions and asking "why?"

Analytical Thinking is the ability to evaluate a situation holistically, break it into key components, synthesize information, and analyze data to make the best decision in work or life.

The **B.A.D** Framework: A Structured Approach to Data Analysis Cases

A repeatable process to crack data business cases, align with industry needs, and showcase technical expertise

Business Context Mastery

→ Understand industry,
stakeholders, and problems

Analytical Strategy Design

→ Structure the problem,
prioritize hypotheses, plan
analysis

Data-Driven Execution

→ Analyze data, validate
hypotheses, communicate
insights.

Business Context Mastery

Define the problem

1 Clarify the Business Problem

- **What is the business objective?** (e.g., increase revenue, reduce costs, improve customer satisfaction).
- **What are the key metrics/KPIs?** (e.g., conversion rate, churn rate, inventory turnover).
- **Who are the stakeholders?** (e.g., marketing team, CFO, operations).
- **What constraints exist?** (e.g., budget, time, data availability).

2 Decode the Industry

- **Identify industry-specific challenges and benchmarks** (e.g., retail: seasonality; healthcare: patient retention).
- Use frameworks like **PESTEL Analysis, Porter's Five Forces, SWOT Analysis, 4P/7P Marketing Mix, 5 Whys**

3 Identify Key Stakeholders

- Example: For a supply chain optimization case, stakeholders might prioritize cost reduction (finance) vs. delivery speed (operations).

1 Clarify the Business Problem

Business Objectives

- **Understand customer behavior**, identify purchasing and engagement patterns.
- **Optimize marketing effectiveness**, increase CTR, CR, and reduce CPA.
- **Improve profitability**, allocate budget efficiently to maximize ROAS.

Key Metrics

- **Ad Performance Metrics** (e.g., Click, Impression, Click-through rate, Conversion rate)
- **Profitability Metrics** (e.g., Ad spend, Profit & Loss)

Stakeholders

- **Marketing Team** – Ad performance, targeting, creative optimization.
- **CMO** - Improve campaign performance.
- **CFO** - Budget optimization, profitability.
- **Sales Team** - Increase conversion and revenue.

Analytical strategy design

Define the problem

1 Break the Problem into Drivers

- Use MECE (Mutually Exclusive, Collectively Exhaustive) to split the problem into components.
- Example: For "Why are sales declining?", split into pricing, competition, product quality, and customer segments.

2 Prioritize Hypotheses

- Identify industry-specific challenges and benchmarks (e.g., retail: seasonality; healthcare: patient retention).
- Use ICE Scoring

3 Design the Analytical Plan

- Descriptive → Trend analysis, comparison, segmentation
- Diagnostic → Funnel analysis, correlation, clustering
- Predictive → Forecasting, ML (optional)

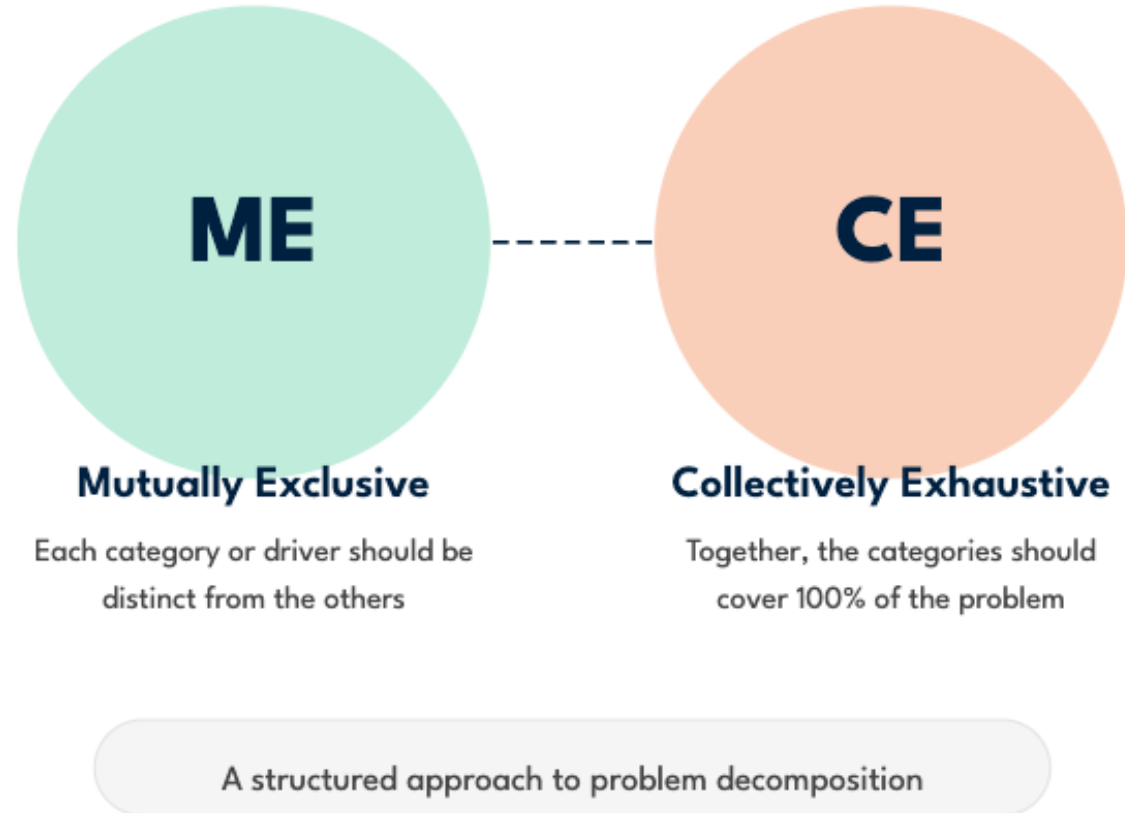
Break the Problem into Drivers using the MECE Principle

Structure the problem using MECE (Mutually Exclusive, Collectively Exhaustive)

Step 4 – Use MECE to Break It Down

Structuring analysis with clear, comprehensive categories

What does MECE mean?



The MECE Principle

Mutually Exclusive, Collectively Exhaustive

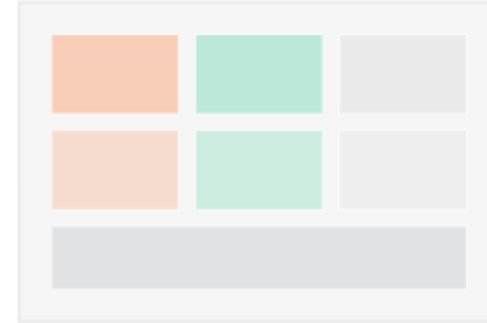
Mutually Exclusive

No Overlap Between Categories



Collectively Exhaustive

Covers All Possibilities



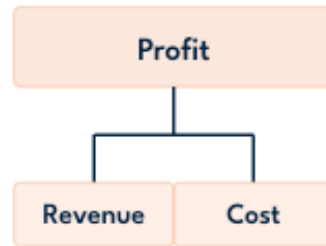
An approach to organizing information that ensures categories don't overlap and together cover all possible options

Three MECE Structuring Techniques

Choosing the right approach for your analysis

Function

Algebra Structure



Mathematical formulas to structure problems

Process

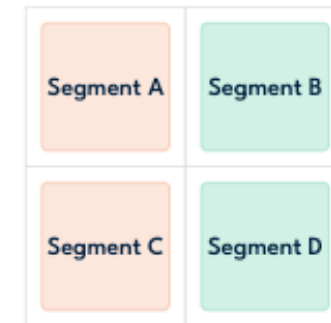
Sequential Steps



Operational sequence breakdown

Segment

Categories

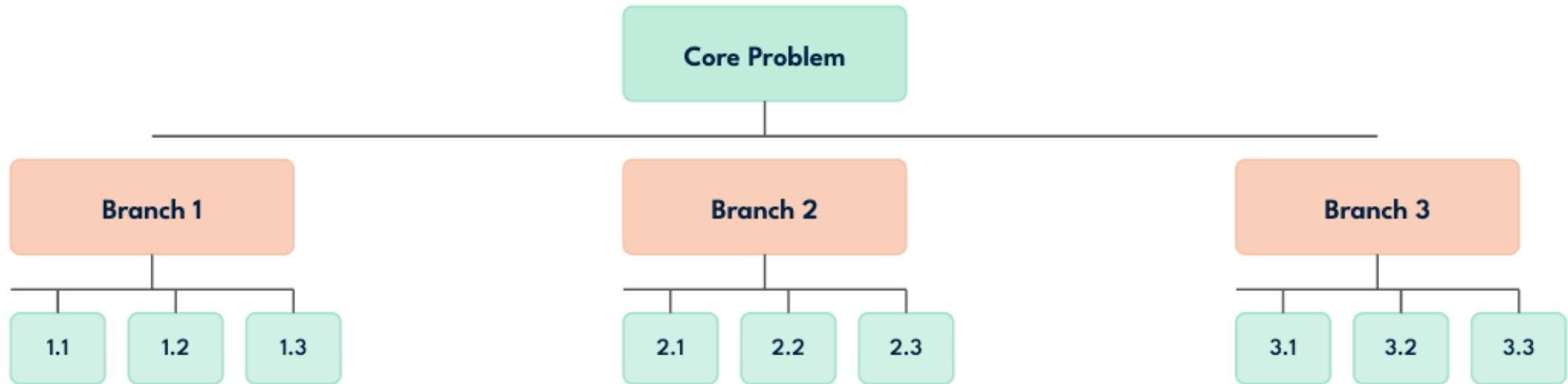


Dividing by customer types, regions, etc.

Choose the technique that best fits the nature of your problem

Issue Tree Visual Structure

Organizing complex problems with hierarchical analysis

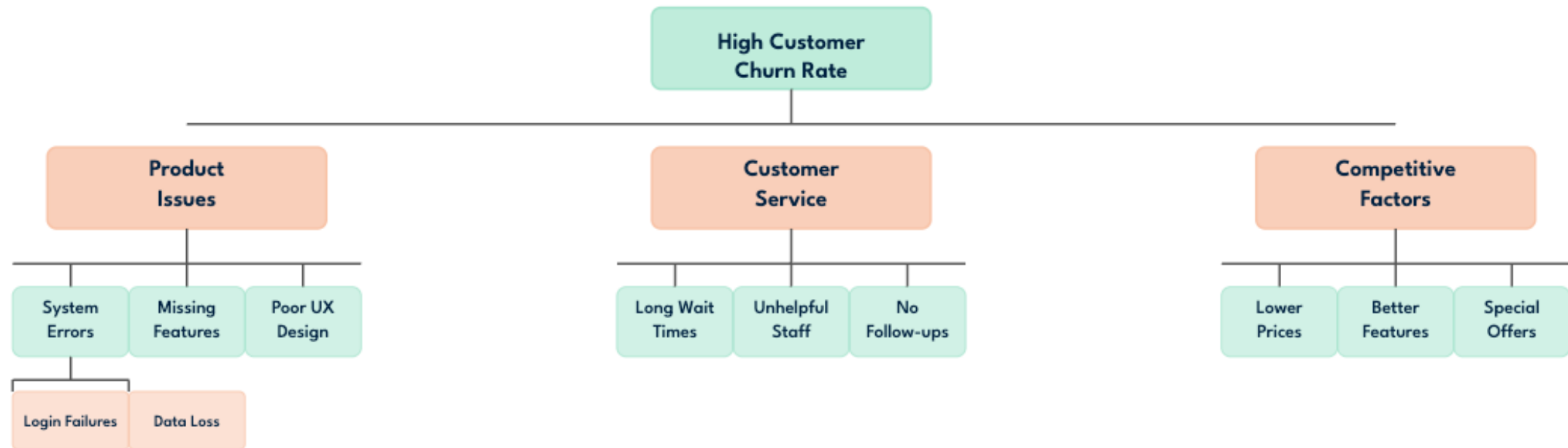


Key Properties of Issue Trees:

- MECE - No overlap between branches, covers all possibilities
- Hierarchical - Break complex problems into smaller, manageable parts

Customer Churn Issue Tree Example

Analyzing root causes of high customer churn rate

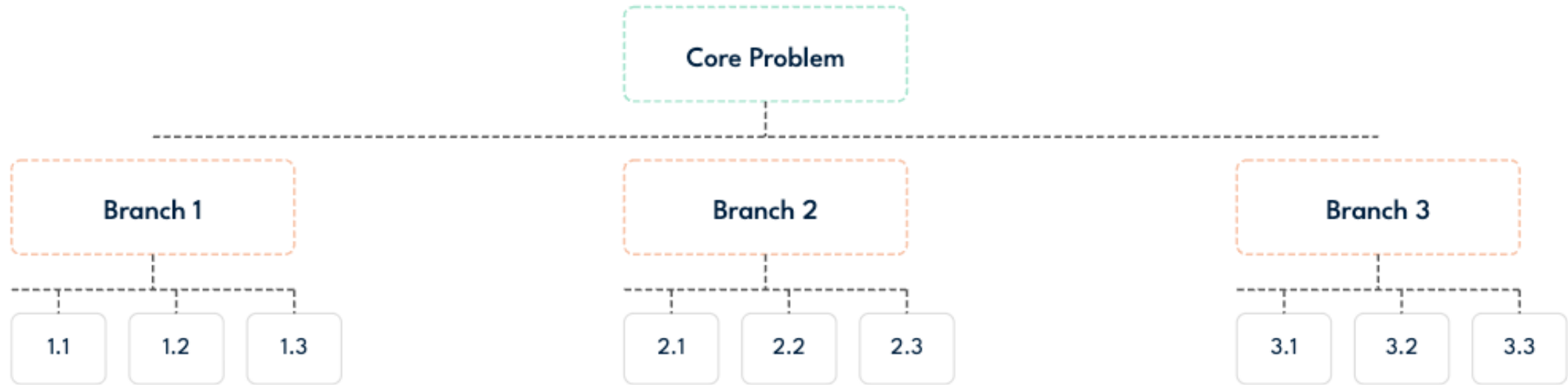


Data-Driven Decision Making:

- Customer survey shows 45% report system errors as primary reason for leaving
- Support ticket data reveals average wait time of 48 minutes (industry avg: 15 min)
- Competitor analysis shows our prices are 20% higher with fewer features

Issue Tree Template

For Student Practice



Student Instructions:

1. Identify a business problem and place it in the Core Problem box
2. Break down the problem into 3-5 MECE branches (main causes)
3. For each branch, identify sub-causes that are MECE
4. Continue breaking down until you reach actionable insights

Math Tree Technique Example

Breaking down metrics with mathematical relationships



Math trees ensure MECE by using mathematical relationships

Breaking Down Problems Step by Step

Using dimensional analysis to create mathematical formulas

Dimensional analysis is a method that helps you break down most mathematical formulas when creating Math Trees. All you need to do is identify the key factors that influence the problem.

How to Use Dimensional Analysis to Transform Metrics into Formulas

Step 1: Identify the metric to break down

Step 2: Find a direct numerical factor

Revenue

Number of Customers

Complete Breakdown Example

Revenue

=

Number of Customers

×

Revenue per Customer

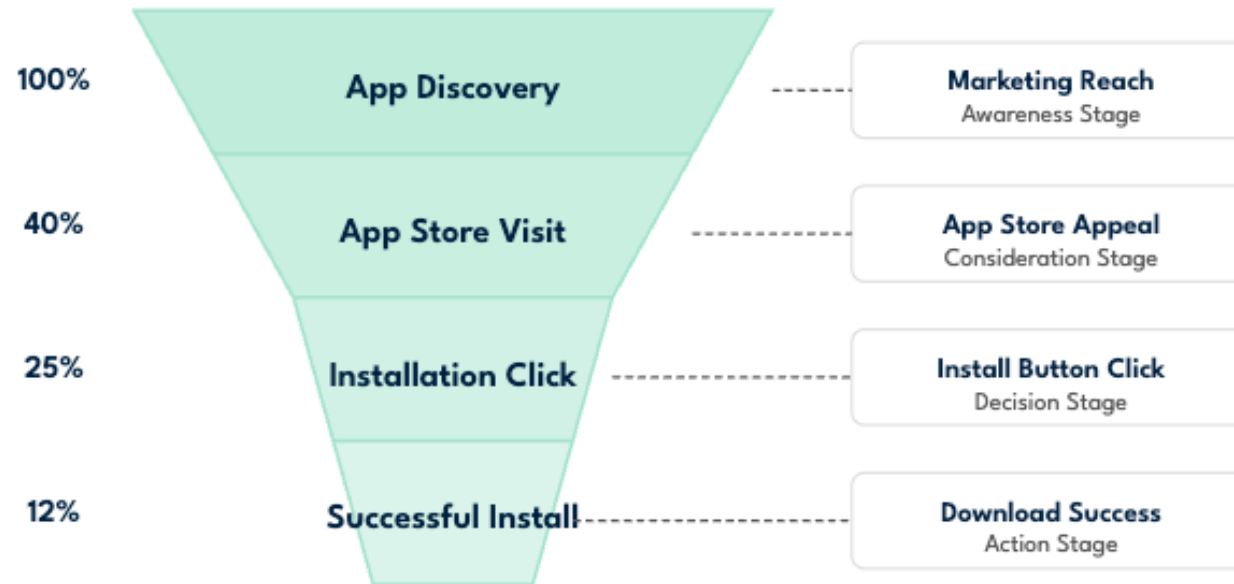
$$\text{Revenue} = \text{Number of Customers} \times (\text{Revenue per Customer})$$

Analysis by Conversion Funnel

Tracking user flow through multi-step processes

The funnel method helps analyze metrics that represent outcomes of a multi-step conversion process.

How to increase mobile app installation rate?



Key Funnel Insights

- Largest drop-off occurs between App Discovery and App Store Visit (60%)
- Only 48% of users who click Install complete the installation process

The Algebraic Approach: What & Why

Breaking down business problems with mathematical formulas

What is it?

The Algebraic Approach breaks down problems into a mathematical formula with variables.

It's commonly used in case interviews for management consulting, management

Why use it?

Nearly all companies use numbers to measure performance.

In case interviews, you'll be asked to improve metrics like profit, revenue, market share, etc.

Examples of Algebraic Formulas

Profits Formula

$$\text{Profits} = \text{Revenues} - \text{Costs}$$

OR

$$\text{Profit} = \text{Revenues} \times \% \text{ Margin}$$

Hiring Costs Formula

$$\text{Hiring costs} = \text{Number of people hired} \times \text{Cost per hire}$$

The Algebraic Approach follows the MECE principle (Mutually Exclusive, Collectively Exhaustive)

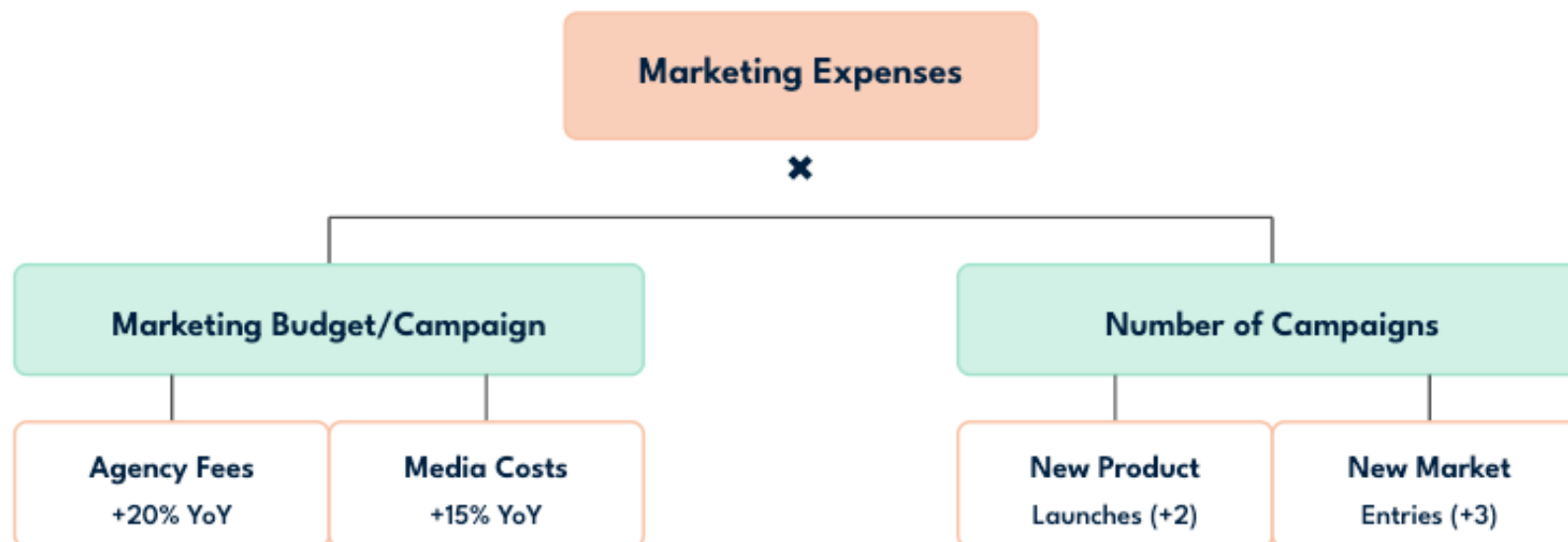
Using the Algebraic Approach Effectively

Best practices for analyzing business problems

Key Considerations:

- Apply the approach **selectively** - focus on the most relevant variables
- Don't just find the formula - evaluate the **real-world impact** of each variable
- Dig deeper to understand **root causes** - why are costs increasing?

Issue Tree: Rising Marketing Expense Analysis



Process Structure Analysis

Benefits and application to business processes

Benefits of Process Structure Analysis:

- **Real-world relevance:** All organizations follow processes daily (production, logistics, maintenance, sales, recruitment)
- **Sequential clarity:** Processes naturally follow a logical order from beginning to end
- **MECE-compliant:** Ensures no aspects of the process are overlooked or duplicated

Visual Example: Sales Process

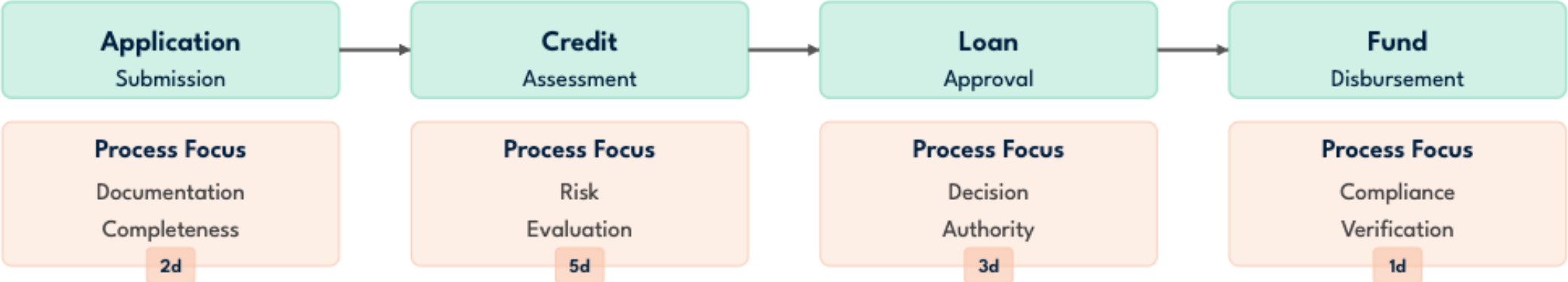


Each step can be further analyzed for potential improvements while maintaining MECE organization

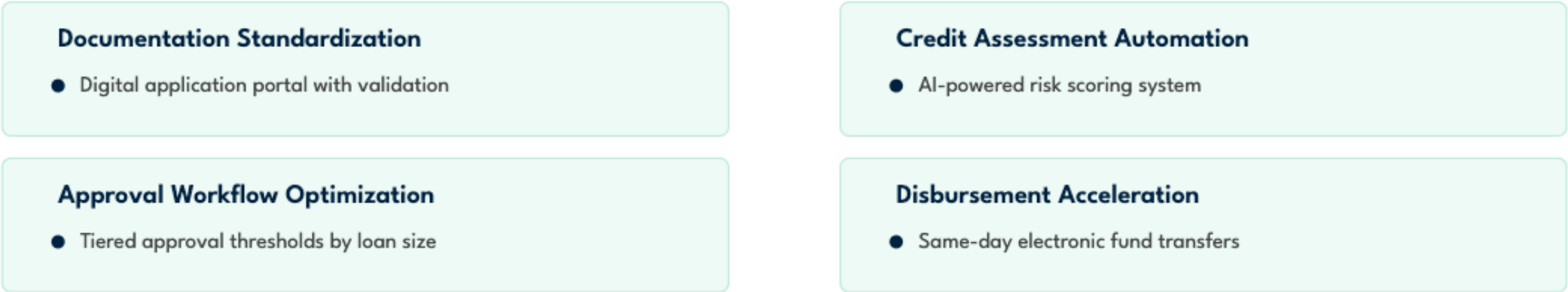
Bank Loan Issuance Process Analysis

Sequential steps and improvement opportunities

Understanding the sequential steps of the bank loan issuance process



Process Improvement Opportunities



Structured Thinking Frameworks

Essential frameworks for business problem analysis

Structured thinking frameworks for organizing qualitative problem analysis

3Cs Framework



Analyzes market position from three perspectives

4Ps Framework



Marketing mix elements for strategy development

Applications & Considerations

Strengths

- Quick structure for complex problems
- Valuable for qualitative analysis

Limitations

- More challenging to ensure MECE
- May need validation with data

Simple MECE Frameworks

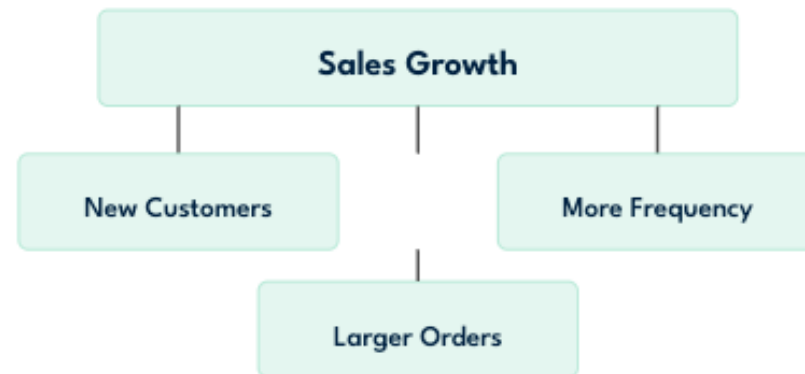
Quick approaches for effective problem categorization

Simple frameworks for quick MECE categorization

Segmentation

Divides problems into distinct categories

Example: Increase Store Sales



Complete coverage, no overlaps

Opposite Words

Creates binary contrasting categories

Example: Sales Growth



Perfect MECE (A or Not A)

When to use: Quick analysis with limited data or time constraints

Hypothesis Prioritization

Using the ICE framework to focus your analysis

"Not all hypotheses are created equal."



Problem:

- Too many possible causes in your MECE issue tree
- Limited time and resources to analyze everything



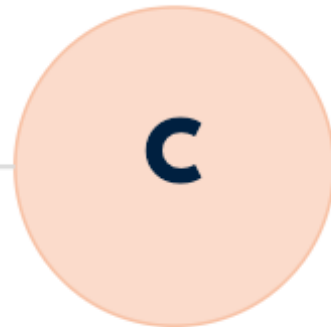
Solution:

Use a scoring framework to decide which hypotheses to test first

Introducing the ICE model:



Impact



Confidence



Ease

The ICE Framework

Practical application for hypothesis prioritization

What is the ICE Framework?

Factor	Meaning	Questions to Ask
 Impact	How big is the potential business effect?	• If this is true, how much will it help solve the problem? • What would be the ROI?
 Confidence	How certain are we that this cause is real?	• Do we have past data or signals that support this idea? • Is there evidence for this hypothesis?
 Ease	How easy is it to test this with our data/tools?	• Can I test this quickly or will it require heavy effort? • Do we have the right data available?

ICE Score = Impact × Confidence × Ease

ICE Example Table

Prioritizing hypotheses for a sales decline problem

Problem: Sales are dropping

Hypothesis	Impact	Confidence	Ease	ICE Score
Low traffic from paid ads	5	4	5	14
Product pricing is too high	4	3	3	10
Bug in checkout page	3	2	2	7

Prioritize analysis on top-scoring hypotheses first

ICE Score = Impact × Confidence × Ease

Hypothesis Types Framework

Asking the right questions for effective analysis

Start From Your Prioritized Hypothesis:

"Users on mobile churn more than desktop users"

 Ask: What kind of question is this?

Descriptive

Is it about patterns?

Diagnostic

Is it about causes?

Predictive

Is it about predicting future behavior?

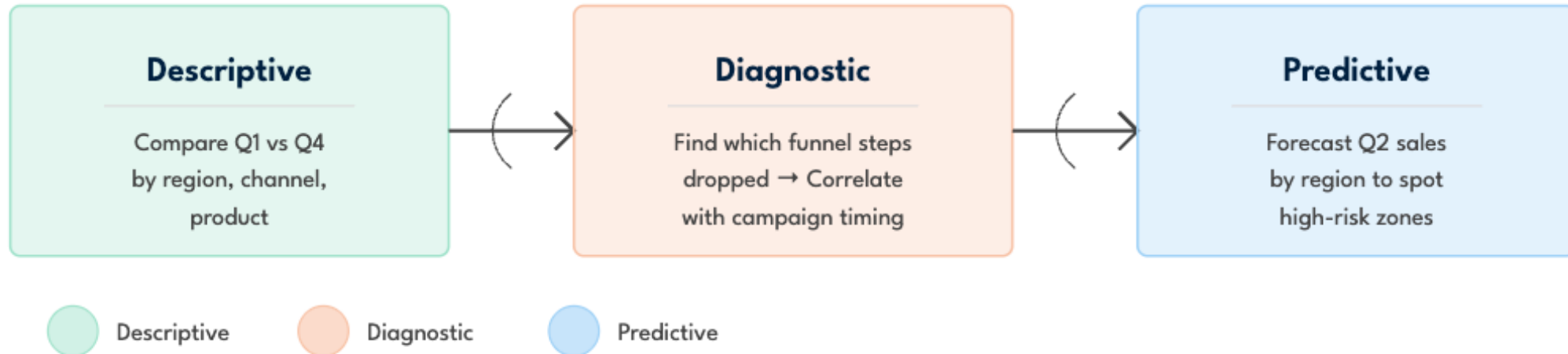
Hypothesis Type	Question	Example	Suggested Techniques
Descriptive	What is happening?	Are churn rates higher for mobile?	• Trend analysis • Segmentation • Group comparison
Diagnostic	Why is it happening?	Why are mobile users churning more?	• Funnel analysis • Correlation • Cohort analysis • Clustering
Predictive	What will happen next?	Which users are likely to churn?	• Forecasting • Regression • Classification • Survival analysis

Case Study: Analysis Flow

Moving from problem identification to action planning

Case: "Sales dropped in Q1"

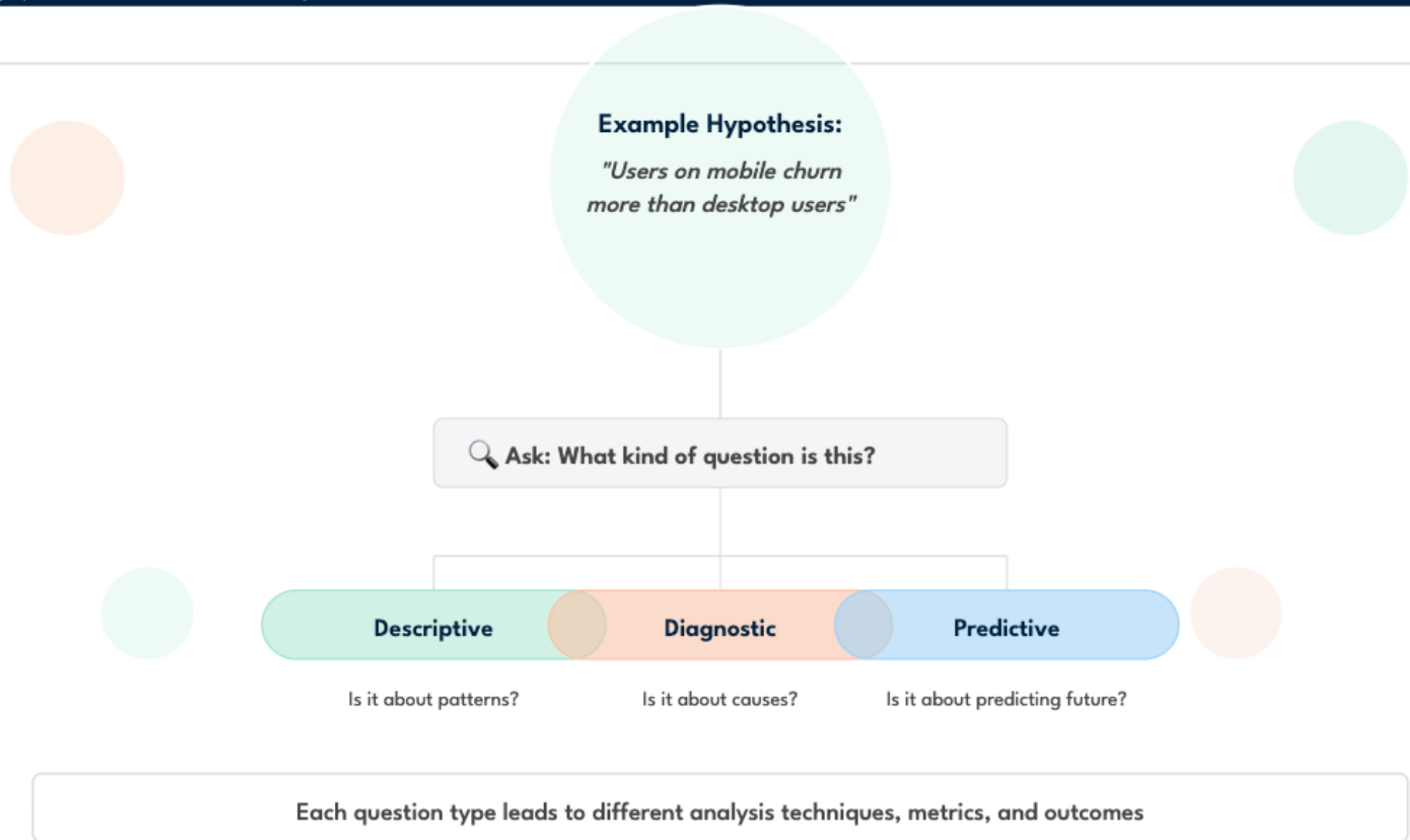
💡 **Tip:** Descriptive helps spot the problem → Diagnostic explains the cause → Predictive helps act ahead.



Use a left-to-right arrow layout to visualize analysis flow

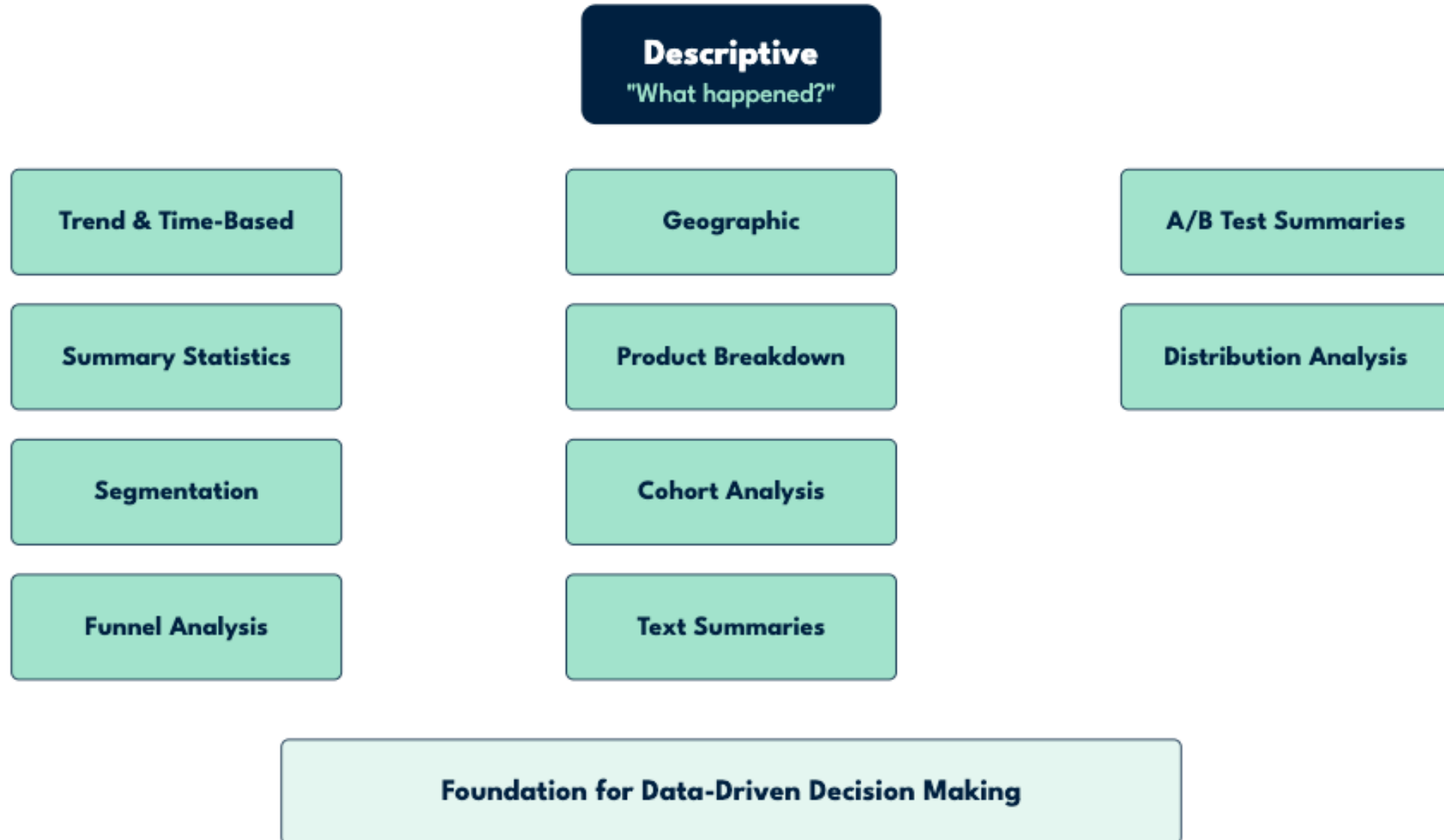
Hypothesis Decision Tree

Categorizing questions for effective analysis



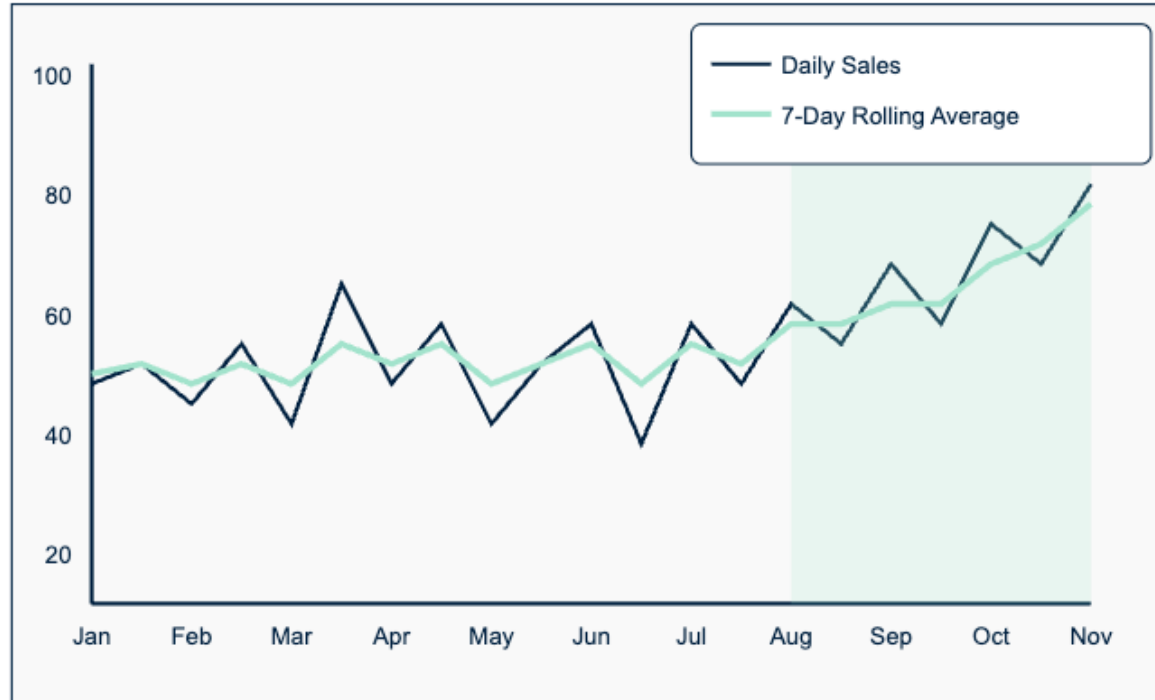
Descriptive Analytics Methods

From Data to Insights



A. Trend & Time-Based Analysis

Understanding how key metrics evolve over time



Key Techniques:

- Line charts (daily/weekly/monthly trends)
- Rolling averages (7-day, 30-day smoothing)
- Period-over-period comparison (WoW, MoM, YoY)
- Seasonality detection & time-series decomposition

Business Application:

An e-commerce company uses time-based analysis to identify:

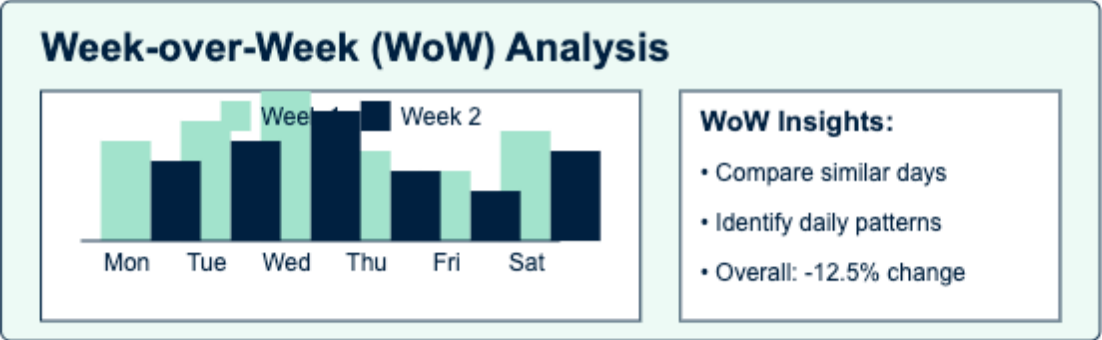
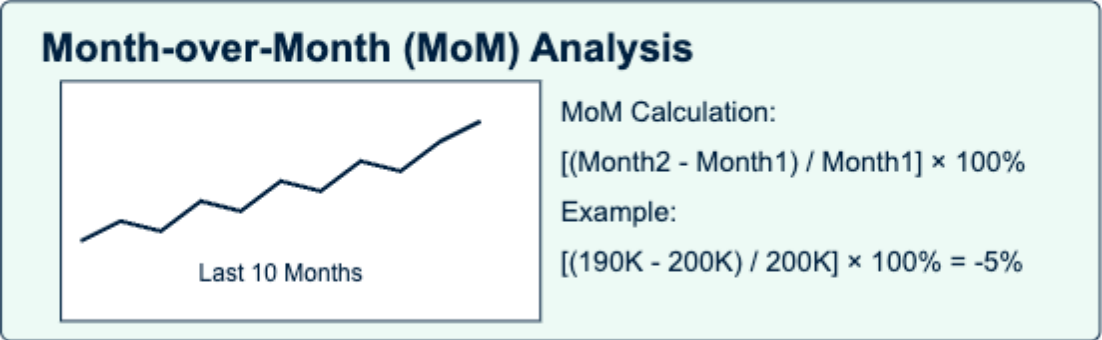
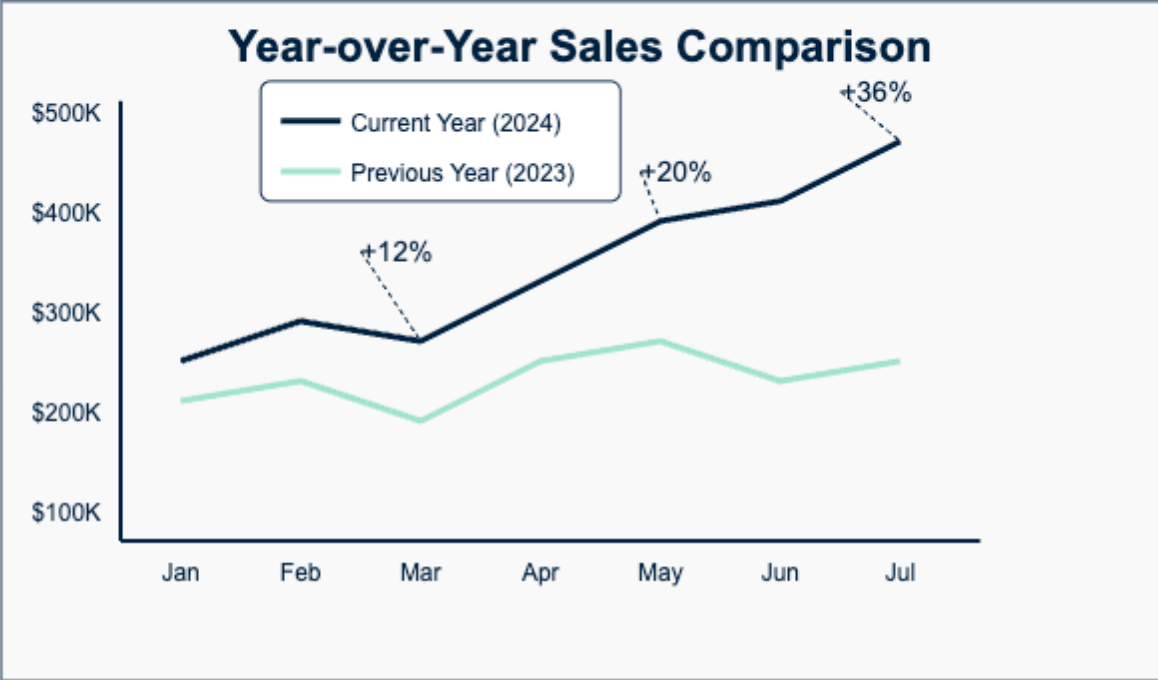
- Sales spikes during holiday seasons
- Day-of-week purchasing patterns
- Year-over-year growth rates

Implementation Tip:

Apply smoothing techniques (like 7-day rolling averages) to filter out noise and identify true patterns in volatile daily data.

Period-over-Period Comparison

Comparing performance across similar time intervals



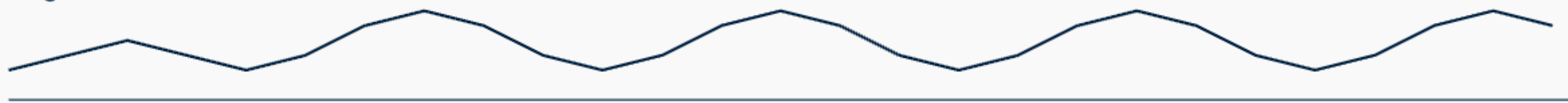
Business Benefits of Period-over-Period Analysis:

- Identifies growth trends and momentum (acceleration or deceleration)
- Reveals cyclical patterns and seasonality effects
- Measures marketing campaign effectiveness over similar time frames
- Normalizes for seasonal variations when comparing YoY same-period data

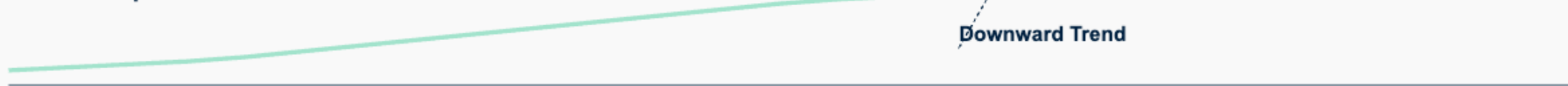
Seasonality Detection & Time-Series Decomposition

Breaking down data into trend, seasonal, and residual components

Original Time Series



Trend Component



Downward Trend

Seasonal Component



Weekly Pattern

Residual Component (Random Variation)



Additive Decomposition:

$$Y(t) = \text{Trend} + \text{Seasonality} + \text{Residual}$$

Business Applications:

- Accurate inventory planning for seasonal demand
- Marketing campaign timing optimization

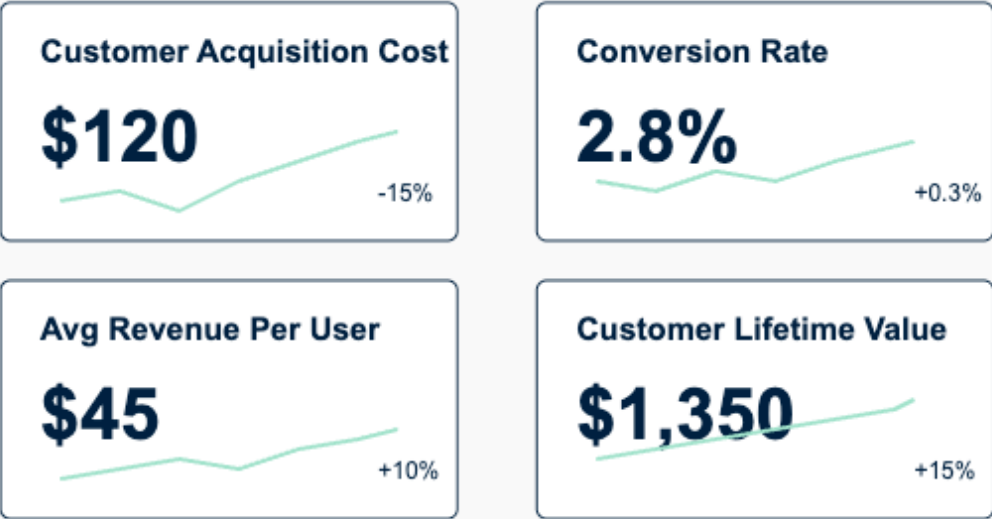
Implementation Notes:

- Common decomposition methods: Additive, Multiplicative
- Tools: Python (statsmodels), R (decompose), Excel (time series functions)

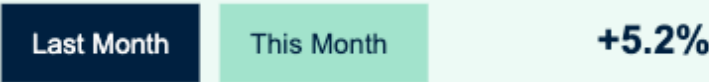
B. Summary Statistics & KPIs

Showing the current state of key business indicators

SaaS Executive Dashboard



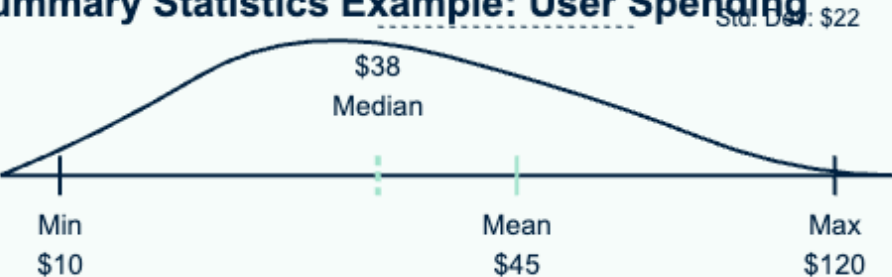
Monthly Growth Rate: +5.2%



Key Techniques:

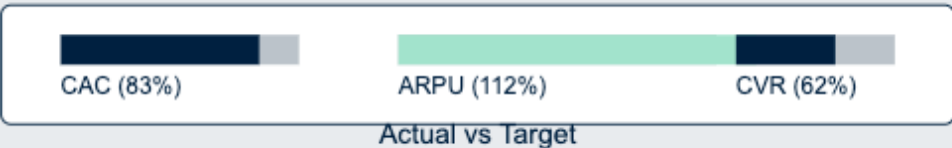
- Mean, median, min, max, standard deviation
- Growth rate (% increase or decrease)
- Cumulative totals
- Ratio metrics (CAC, CVR, AOV, ARPU)
- KPI scorecards & dashboards

Summary Statistics Example: User Spending



Implementation Tip:

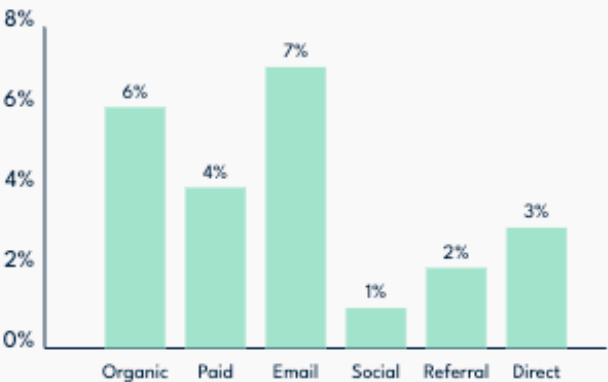
Design KPI dashboards that compare current metrics against targets and historical performance to provide context.



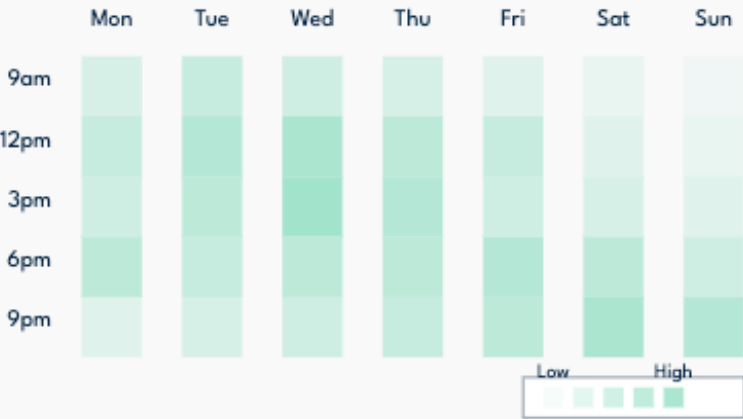
C. Segmentation & Grouped Analysis

Comparing how metrics differ across categories

Conversion Rates by Traffic Source



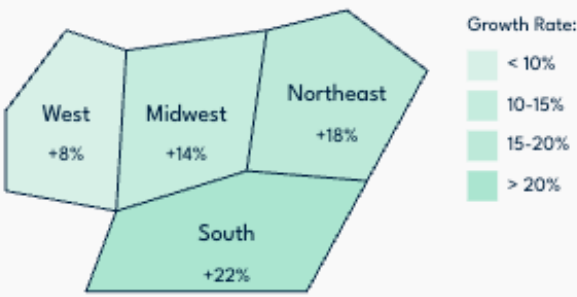
Click-Through Rate by Day & Hour



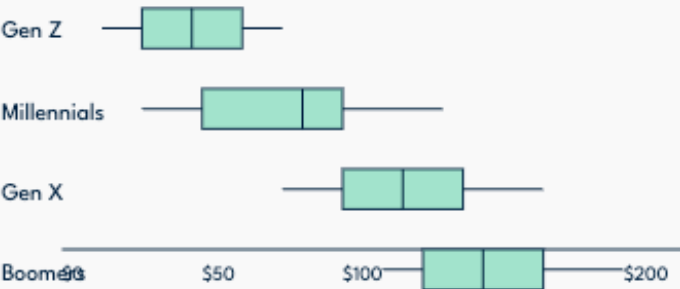
Key Techniques:

- Group-by analysis
- Bar charts (total/average per group)
- Pivot tables (cross-tabulated)
- Box plots (distributions)
- Heatmaps (multi-dimensional data)

Regional Performance Differences



Customer Spend by Demographic



Business Application:

- Conversion rates by traffic source
- ROI comparison across channels
- Customer spend by demographics
- Regional performance differences

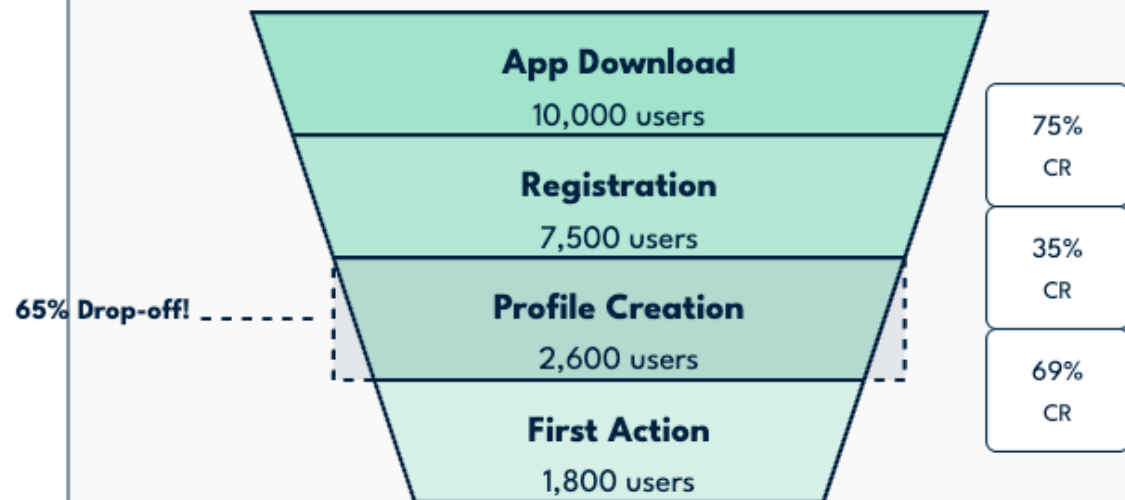
Implementation Tip:

Start with broad segments, then drill down into sub-segments to uncover more nuanced insights.

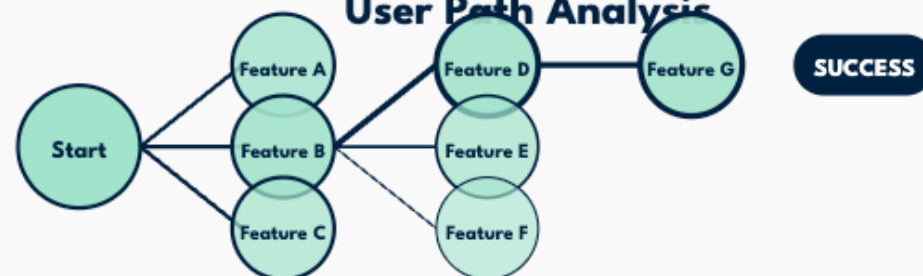
D. Funnel & Path Analysis

Measuring performance across step-based journeys

App User Onboarding Funnel



User Path Analysis



Most Successful User Path:

Start → Feature B → Feature D → Feature G

- 28% of all successful users follow this path
- 3.2x higher retention rate vs. other paths

Key Techniques:

- Funnel charts (Impressions → Clicks → Signups → Purchases)
- Conversion rates at each step
- Drop-off rate analysis
- Path analysis (most common sequences of user behavior)

Implementation Tip:

Focus optimization efforts on steps with the highest drop-off rates to maximize overall funnel performance.

Example: Simplify profile creation by reducing required fields to address the 65% drop-off at this stage.

E. Cohort & Retention Analysis

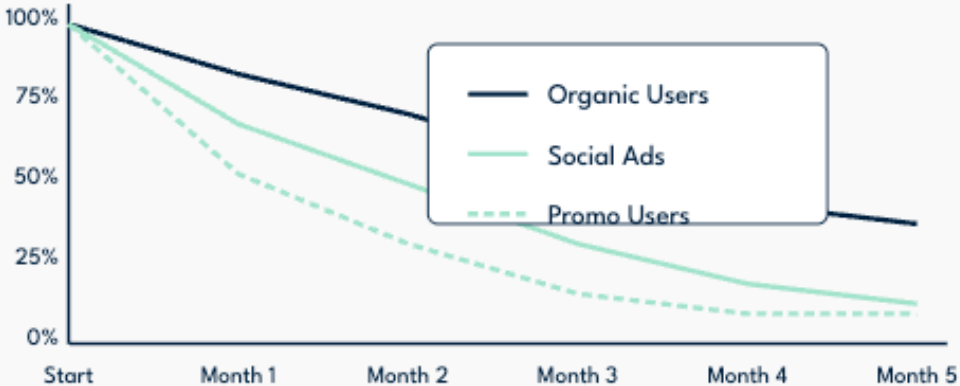
Tracking groups of users over time

User Retention by Signup Month

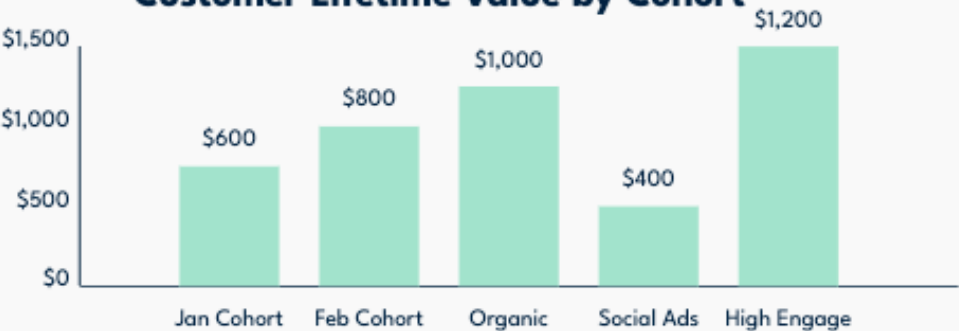
Cohort	M0	M1	M2	M3	M4	M5
Jan 2024	100%	82%	65%	48%	37%	32%
Feb 2024	100%	80%	62%	45%	33%	-
Promo Users	100%	68%	42%	30%	-	-
Organic Users	100%	84%	72%	56%	-	-

Retention % by Month Since Signup

Retention Curves by Acquisition Channel



Customer Lifetime Value by Cohort



Key Techniques:

- Cohort tables (user retention per signup period)
- Retention curves (active users over time)
- Cumulative retention/engagement
- CLTV by cohort

Business Application:

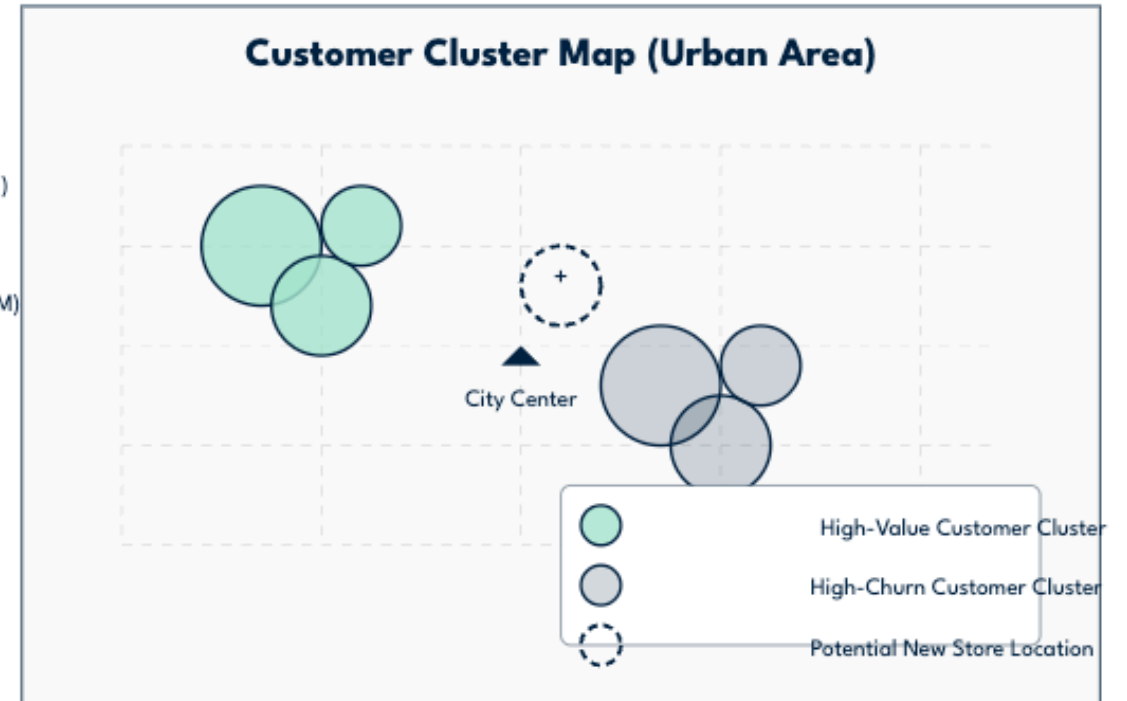
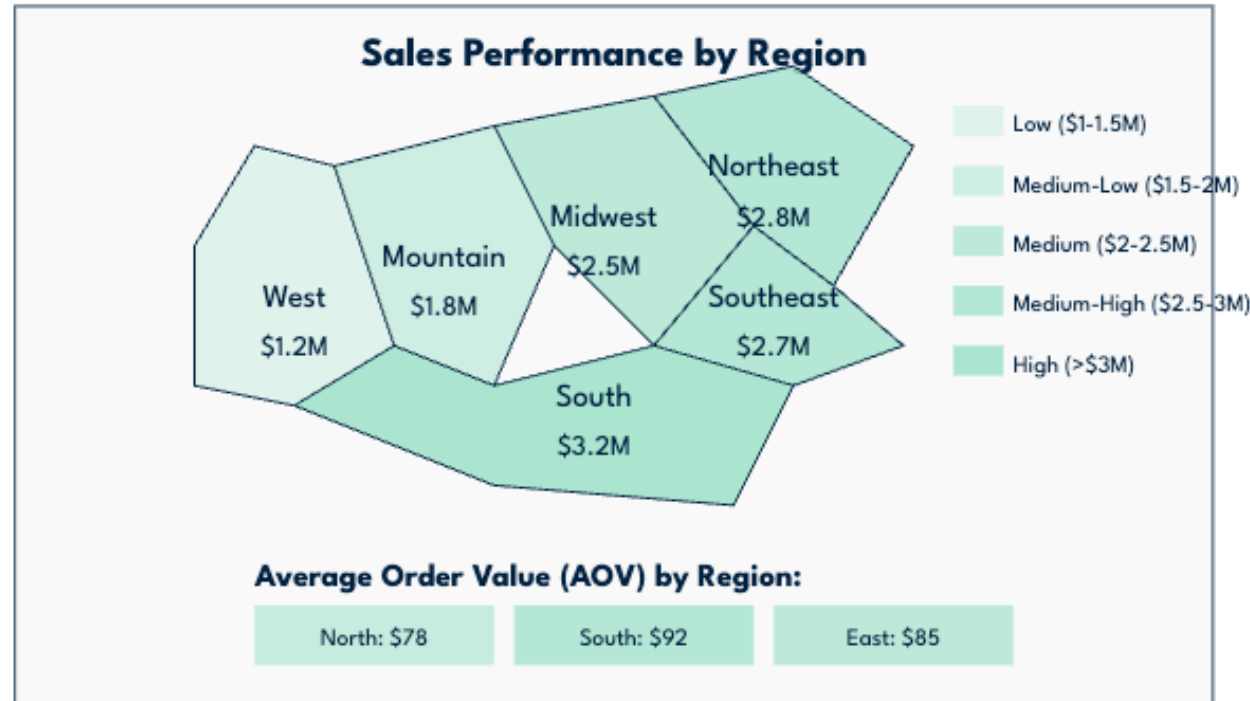
- Promotional vs organic retention
- Impact of product updates
- Highest-value acquisition channels
- Early engagement correlation with long-term retention

Implementation Tip:

Compare cohorts based on common attributes to identify factors that influence long-term value.

F. Geographic & Spatial Analysis

Understanding where performance varies by location



Key Techniques:

- Geo heatmaps (sales by city or country)
- Regional comparisons (AOV in north vs. south)
- Cluster mapping (high-churn zones)

Business Application:

A retail chain uses geographic analysis to:

- Identify high-potential locations for new stores
- Optimize regional marketing spend
- Adjust inventory based on local preferences

Implementation Tip:

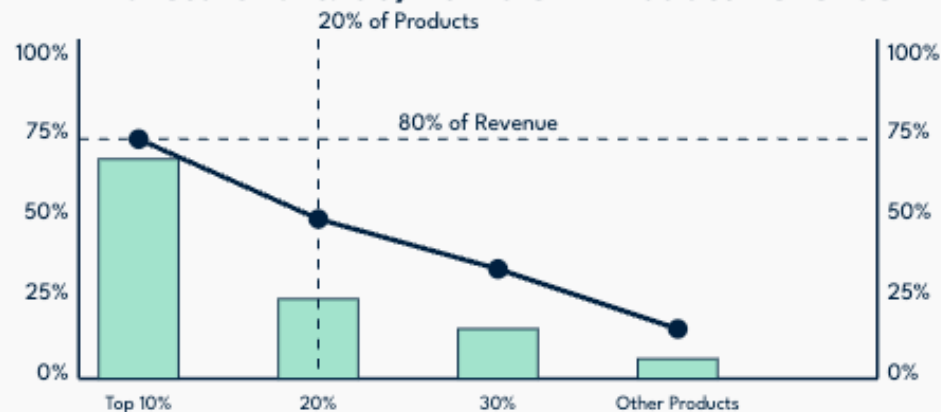
Combine geographic data with demographic information to understand the "why" behind regional performance differences.

Example: Higher-income areas → Higher AOV, Lower frequency

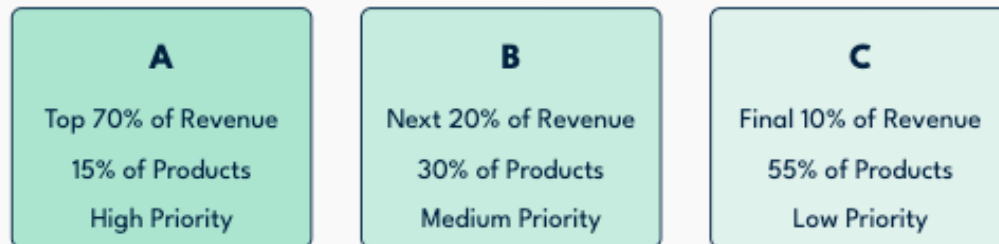
G. Product or SKU-Level Breakdown

Understanding which items drive performance

Pareto Chart: 80/20 Rule in Product Revenue

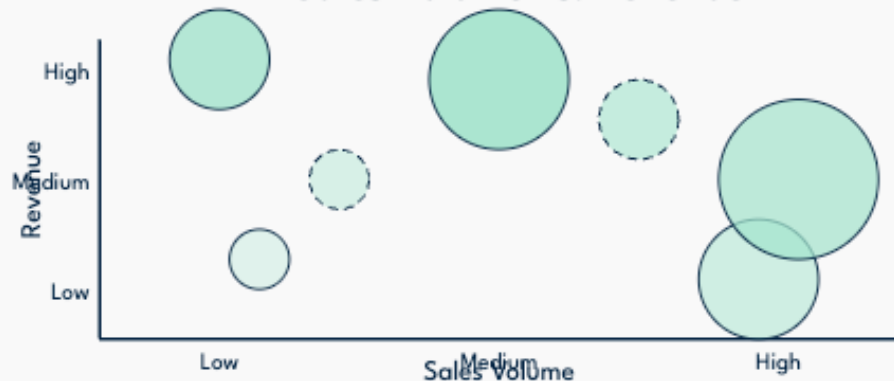


ABC Analysis: Product Segmentation



Category A: High inventory availability, premium shelf space, marketing focus
Category C: Consider discontinuing low performers, minimal stocking

Sales Volume vs. Revenue



Business Application:

A consumer goods company discovers:

- 15% of products generate 70% of profits
- High-volume items with low margins vs. low-volume with high margins
- Complementary product purchasing patterns
- Seasonal variation in product popularity

Key Techniques:

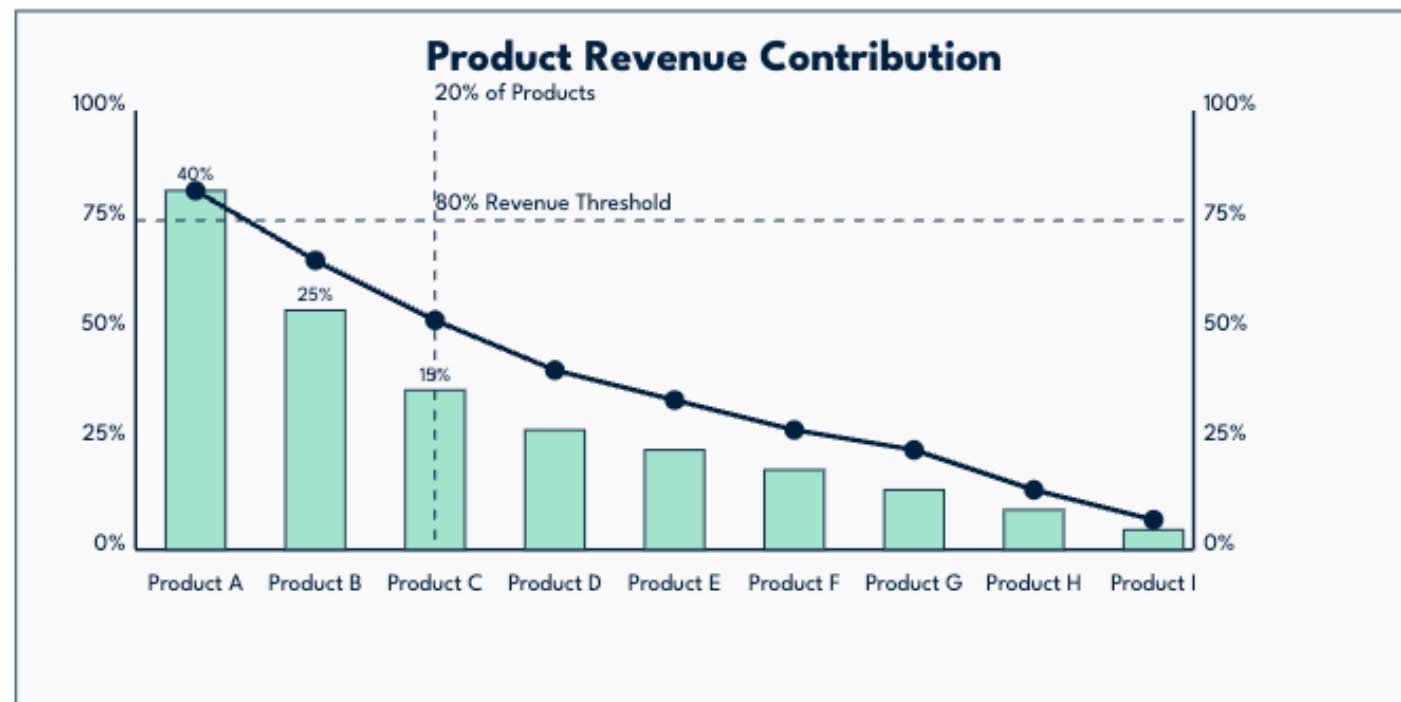
- Pareto charts (80/20 principle)
- ABC analysis (segment by revenue)
- Sales volume vs. revenue bubble charts
- Top/bottom performer tables

Implementation Tip:

Use ABC analysis to categorize products for inventory and marketing prioritization.

Pareto Chart

Visualizing the 80/20 Rule in Business Data



Understanding Pareto Charts

The 80/20 Principle:

- Roughly 80% of effects come from 20% of causes
- Named after economist Vilfredo Pareto
- Common pattern across many business domains

Key Components:

- Bar chart: Individual contribution (left Y-axis)
- Line chart: Cumulative percentage (right Y-axis)
- Items arranged from highest to lowest contribution
- Critical thresholds marked (typically 80% line)
- Focus on "vital few" vs "trivial many"

Business Applications

Product Management

- Identify top revenue-generating products
- Focus marketing on high-impact items
- Rationalize low-performing SKUs
- Prioritize inventory management

Quality Improvement

- Identify most common defect types
- Focus improvement on highest-impact problems first
- Track progress in reduction over time

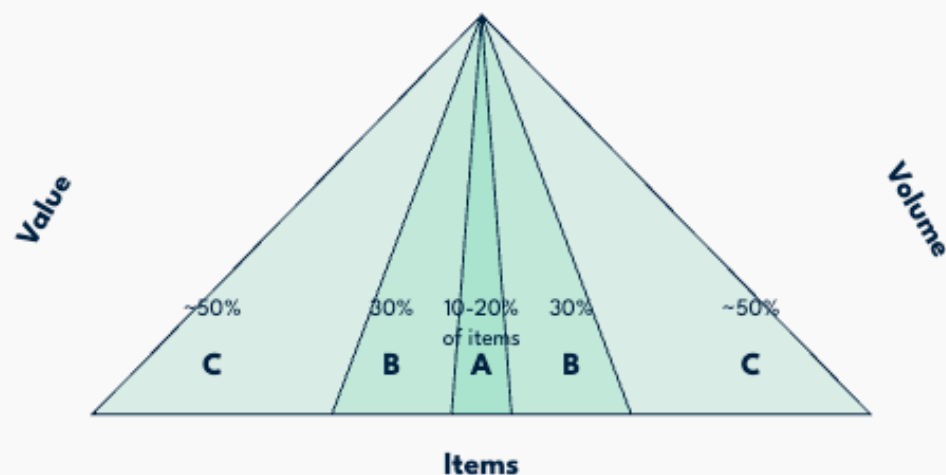
Customer Analysis

- Segment VIP customers (top 20%)
- Identify high-value customer behaviors
- Create tiered loyalty programs
- Develop targeted retention strategies

ABC Analysis

Segmenting Inventory by Value Contribution

ABC Inventory Classification



ABC Categories in Detail

A Items

Very Important
~70% of Value
10-20% of Items
Highest Control
Daily Monitoring
High Availability
Strategic Focus

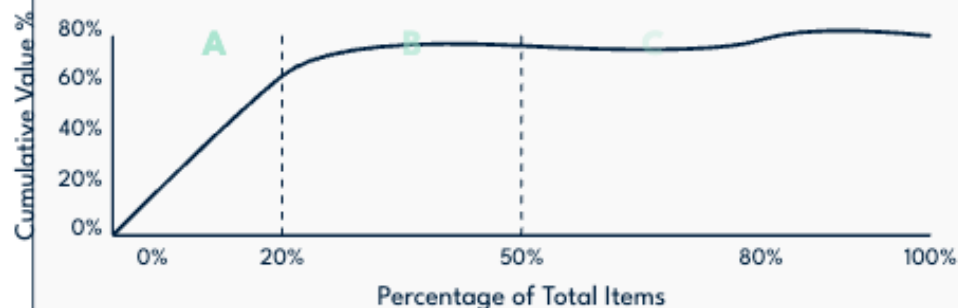
B Items

Moderately Important
~20% of Value
~30% of Items
Medium Control
Weekly Monitoring
Moderate Stock
Tactical Focus

C Items

Least Important
~10% of Value
~50% of Items
Minimal Control
Monthly Review
Limited Stock
Efficiency Focus

Value Distribution Curve



Implementation Best Practices

1. Data Collection & Analysis

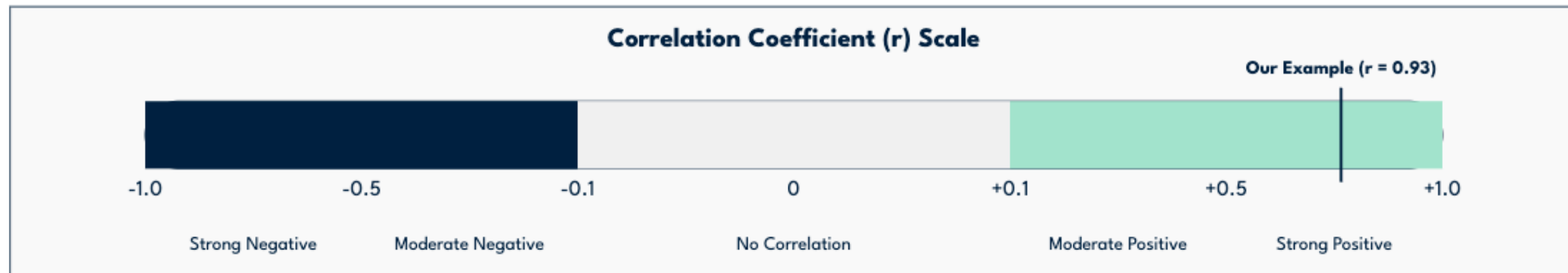
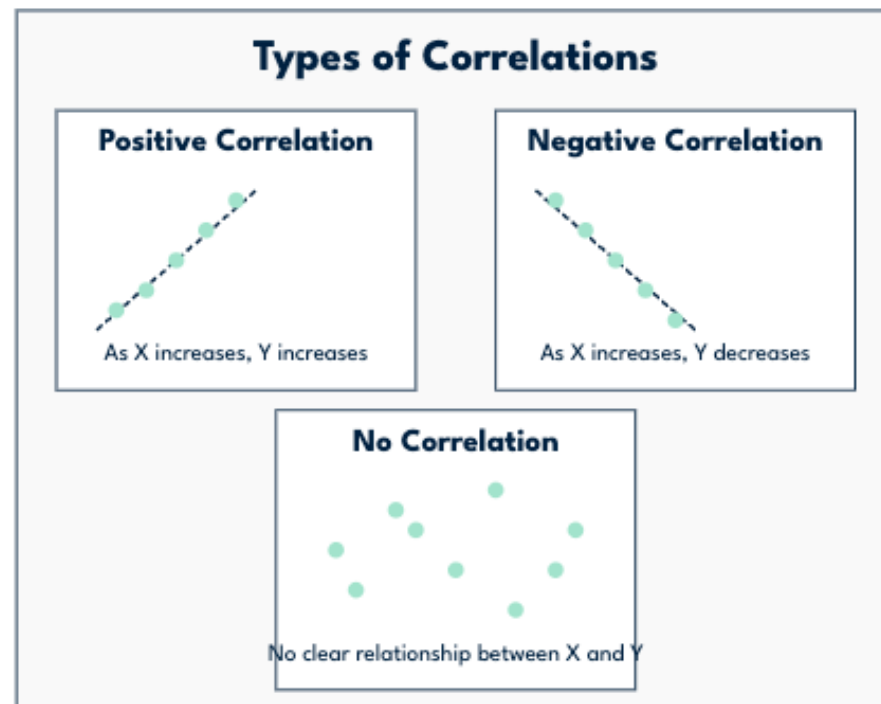
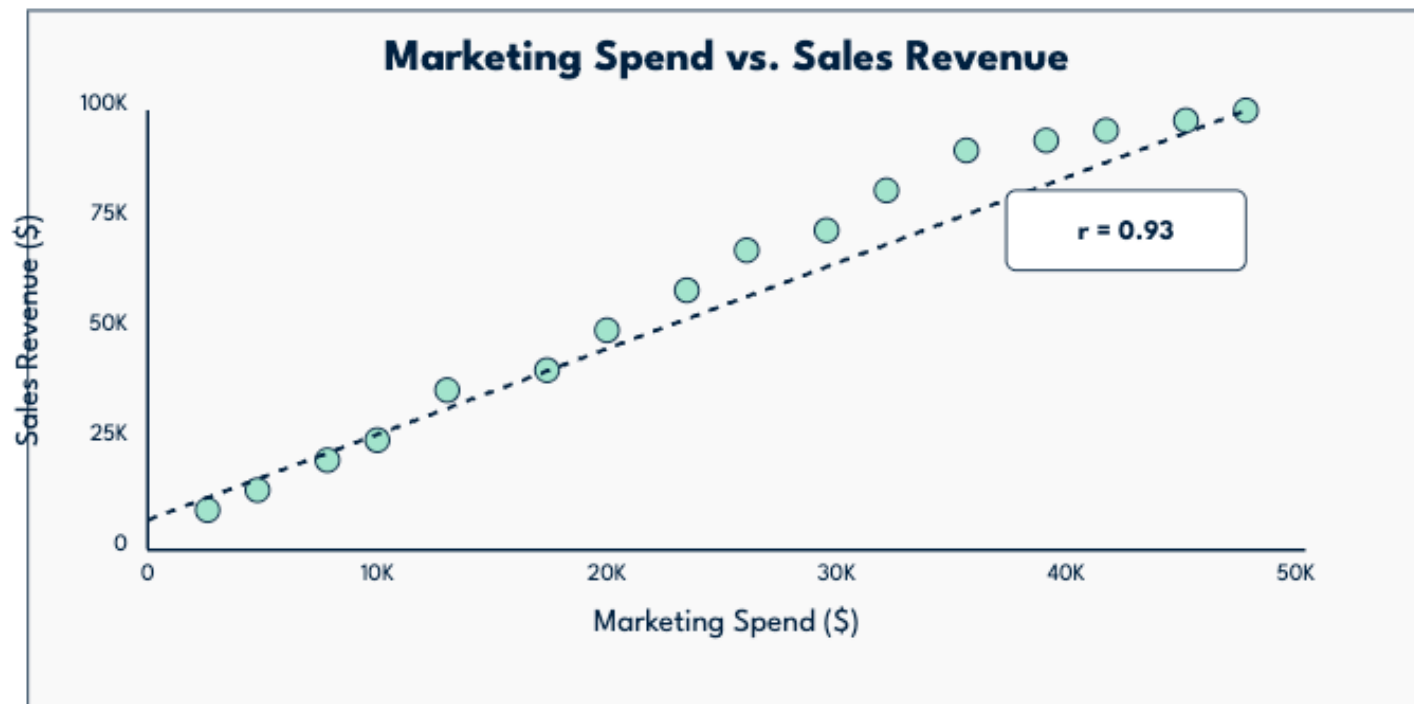
- Collect annual consumption value for each inventory item
- Arrange items in descending order of consumption value
- Calculate cumulative value and percentage of total items

2. Management Strategy by Category

- Customize inventory policies based on classification (A, B, or C)
- Focus resources on managing high-value, high-impact items

Correlation Analysis with Scatterplots

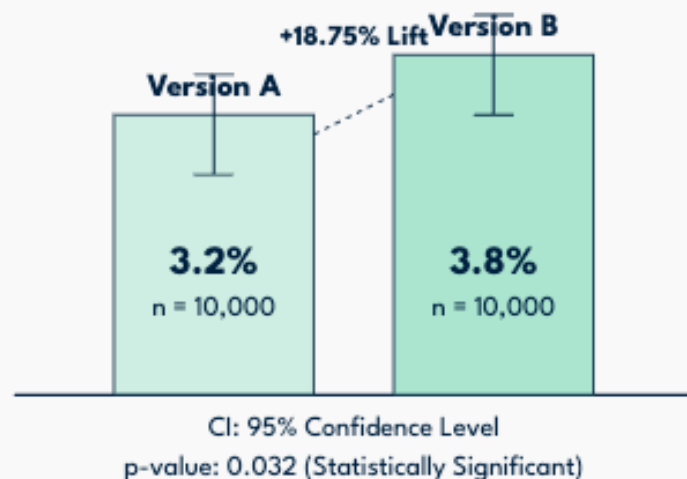
Visualizing Relationships Between Two Variables



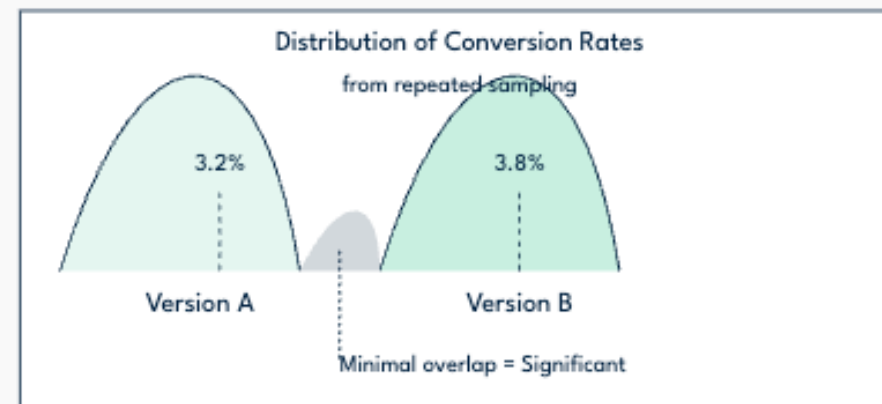
H. A/B Test Summaries

Seeing what happened in each test group

Product Page Conversion Rates



Understanding Statistical Significance



- p-value < 0.05 indicates less than 5% chance the result is due to random variation
- Minimal overlap between distributions suggests real difference between variants

Key Techniques

- Group-wise KPIs (conversion rate A vs. B)
- Lift calculation (% improvement)
- Difference charts with confidence intervals
- Statistical significance testing (p-values)

Implementation Tip

Balance Both Perspectives:

Statistical Significance

Is the difference real?

$$p < 0.05$$

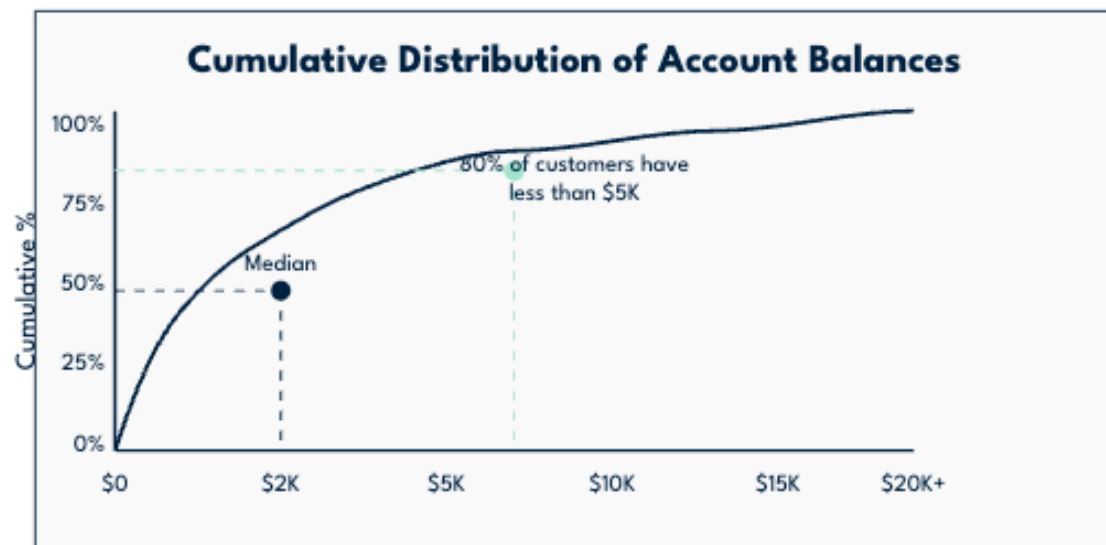
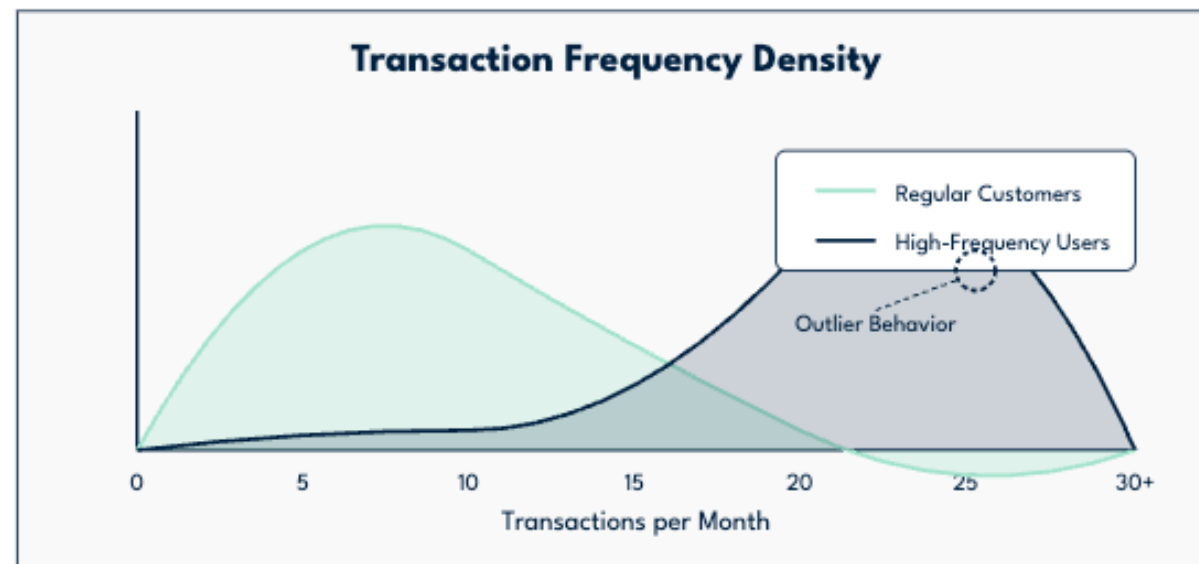
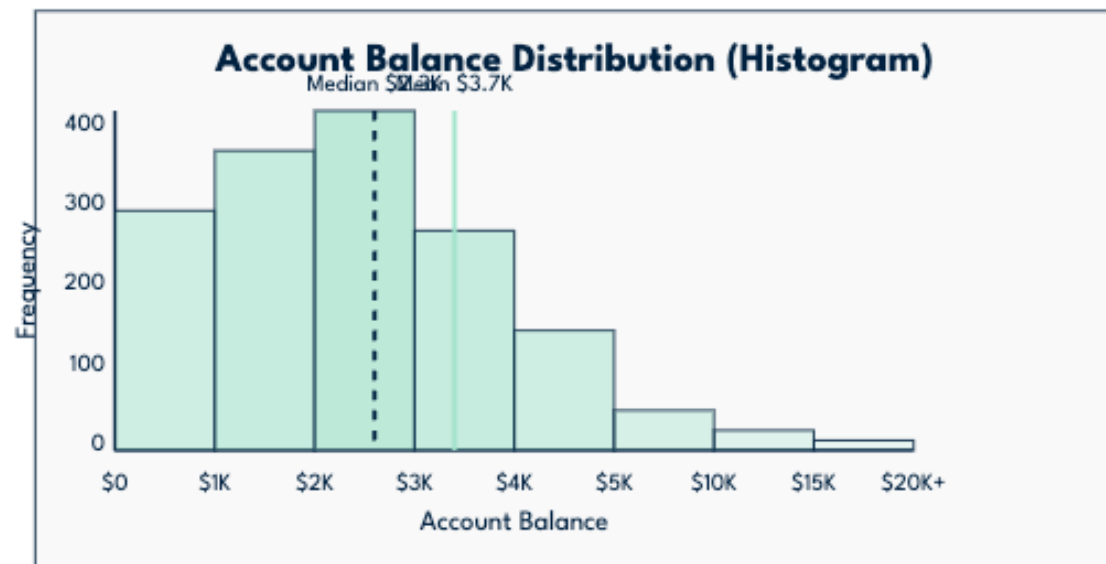
Business Significance

Is the difference meaningful?

ROI positive

I. Distribution & Frequency Analysis

Understanding how values are spread



Key Techniques

- Histograms (number of customers by order frequency)
- Density plots (smooth distributions)
- Frequency tables (counts per bin)
- Cumulative distribution curves
- Distribution statistics (mean, median, mode, variance)

Business Application

A financial services company analyzes:

- Account balance distribution
- Transaction frequency
- Outlier behavior identification
- Customer segmentation

Implementation Tip

Look beyond averages - outliers and distribution shape often reveal critical business insights.

J. Descriptive Text or Survey Summaries

Summarizing qualitative feedback

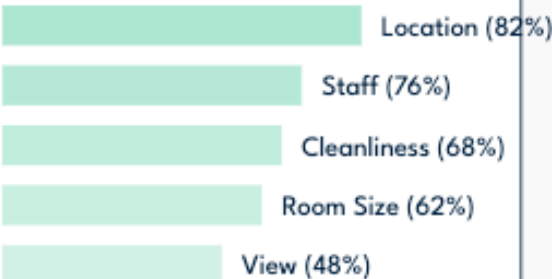
Guest Review Word Cloud



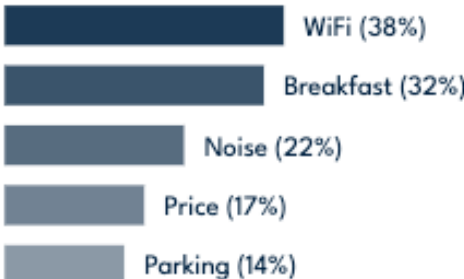
Positive Themes Negative Themes

Top Comment Themes Analysis

Positive Themes



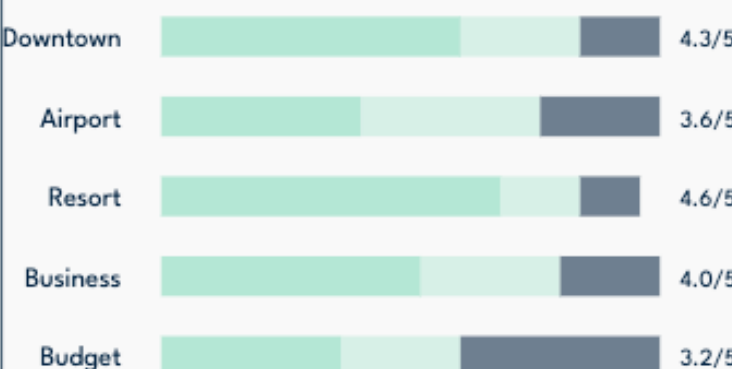
Negative Themes



Sentiment Trends Over Time



Feedback Comparison Across Properties



Key Techniques

- Word clouds (visual summaries)
- Theme analysis (keyword tagging)
- Sentiment scoring (pos/neg/neutral)

Positive Comments
Neutral Comments
Negative Comments

Bringing It All Together

An Integrated Approach to Descriptive Analytics

Descriptive Analytics Framework

1. Summary Statistics

Understand overall performance (KPIs, metrics)



2. Time-Based Analysis

Identify trends and patterns over time



3. Segmentation Analysis

Discover variations across different groups



4. Specialized Analysis

Apply specific techniques for deeper insights



5. Combine Quantitative + Qualitative Insights

Keys to Success



Begin with Clear Business Questions

Choose Appropriate Visualizations

Maintain Data Quality and Consistency

Provide Context for Interpretation

Connect Insights to Actionable Steps

Descriptive Analytics Toolkit

Trend & Time-Based Analysis

Summary Statistics & KPIs

Segmentation & Grouping

Cohort & Retention Analysis

Funnel & Path Analysis

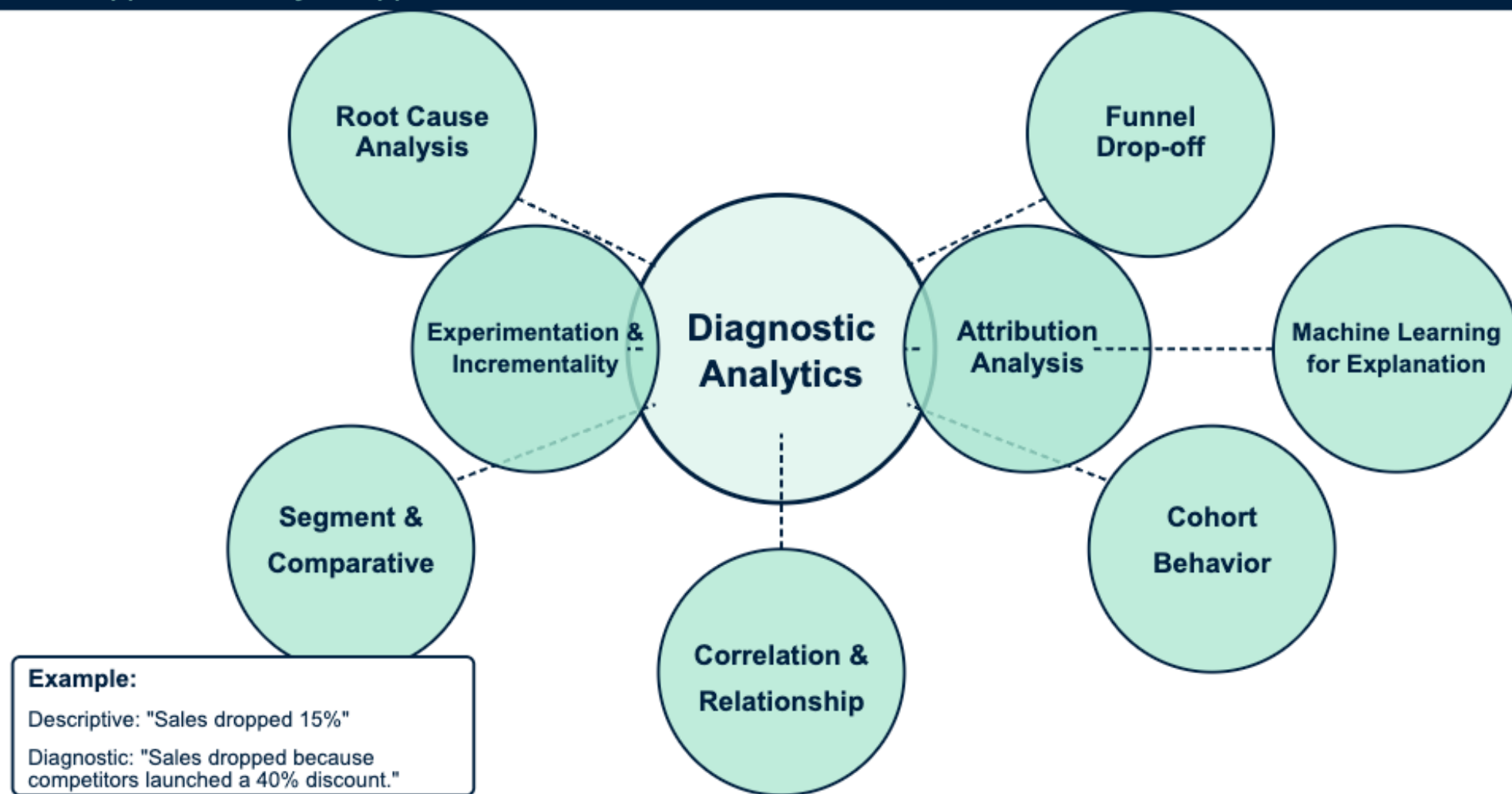
Geographic & Spatial Analysis

Product/SKU Breakdown

Text & Survey Analysis

Diagnostic Analytics: Finding the "Why"

Beyond what happened to why it happened

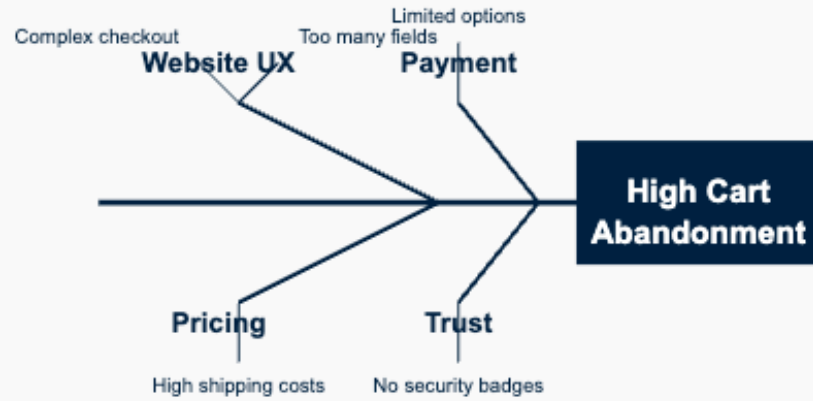


"Descriptive tells you WHAT happened. Diagnostic tells you WHY it happened."

A. Root Cause & Breakdown Analysis

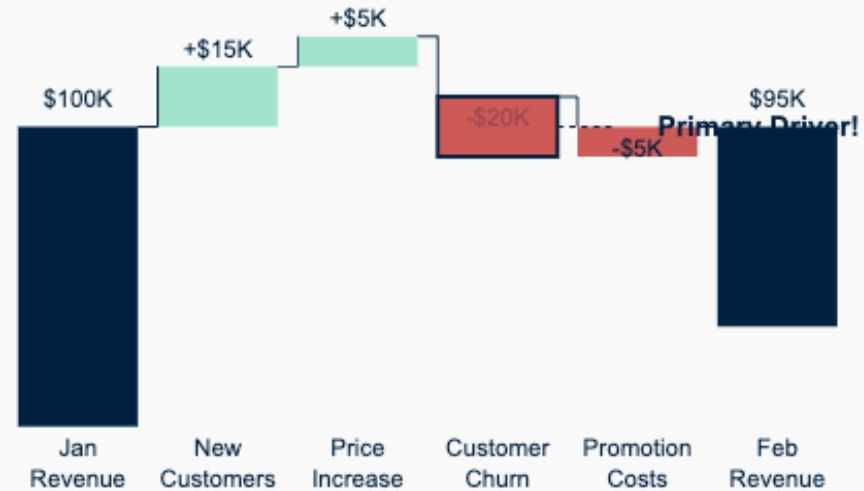
Understanding which factors are driving the problem

Example: Ishikawa (Fishbone) Diagram



Example: Waterfall Chart

Breaking Down Revenue Change (Jan to Feb)



Example: 5 Whys Method

Problem: "Increasing customer service complaints"

Why are complaints increasing? → Response times are too slow

Why are response times slow? → Too few support agents

Why too few agents? → Recent budget cuts reduced staffing

Why cut budget? → Pressure to improve quarterly margins

ROOT CAUSE: Short-term financial goals over customer satisfaction

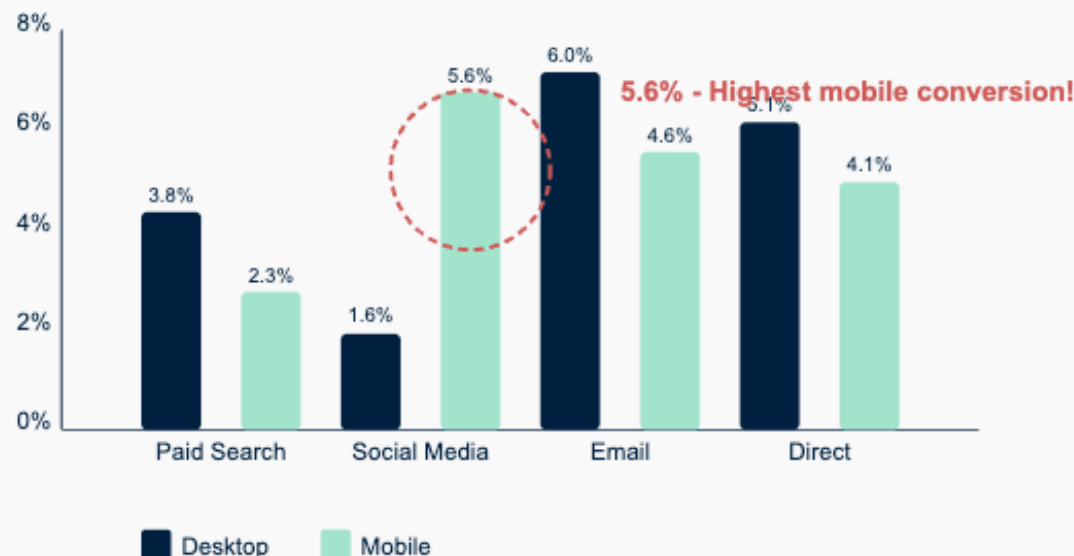
Key Diagnostic Techniques:

- Contribution analysis (% drop in sales by category)
- Drill-down analysis (total churn → by age → by product)
- Root Cause Tree (5 Whys technique)
- Ishikawa (Fishbone) diagram
- Waterfall charts (breakdown revenue changes)

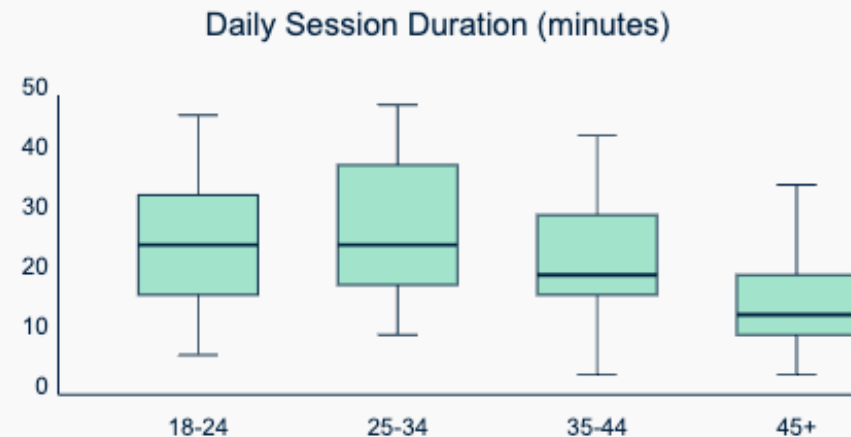
B. Segment & Comparative Analysis

Detect differences across groups that may explain performance gaps

Example: Conversion Rate by Channel & Device



Example: Box Plots - User Engagement by Segment



* Statistically significant difference ($p < 0.05$)

Key Techniques:

- Group comparison charts (conversion rates, engagement by segments)
- T-test/ANOVA (testing if group differences are statistically significant)
- Boxplots across groups (distribution comparison)
- Chi-square test (for categorical outcome differences)
- Relative risk/lift calculation (likelihood to churn by segment)

Business Application:

- Channel Optimization: Allocate budget to highest-performing segments
- Product Strategy: Tailor features to segments with specific needs
- Risk Management: Identify at-risk segments for focused retention
- Personalization: Create targeted experiences for specific groups

T-test Analysis

Testing if differences between two groups are statistically significant

Understanding T-tests

H_0 : No difference between groups ($\mu_a = \mu_b$)

H_1 : Groups are different ($\mu_a \neq \mu_b$)



Group A

Mean = 62.4



Group B

Mean = 73.9

Types of T-tests:

- Independent samples
(Different people in each group)
- Paired samples
(Same people, before/after)

t-statistic = 3.42, p-value = 0.0014 ($p < 0.05$)

Business Case: Email Campaign Test

Email A (Control)



CALL TO ACTION

Email B (New Subject Line)



CALL TO ACTION

Conclusion:

New subject line is **significantly better!**

Group A Results

62.4% open rate
n = 500 recipients

Group B Results

73.9% open rate
n = 500 recipients

When to Use T-tests:

Technical Criteria

- Comparing exactly two groups
- Continuous variable (e.g., revenue, time)
- Approximately normal distribution
- Typically $n > 30$ in each group
- Random sampling/assignment

Business Applications

- A/B tests of marketing campaigns
- Before/after analysis of an intervention
- Comparing performance across segments
- Product version comparisons
- Testing pricing strategy effectiveness

Important Concepts

- p-value: Probability of seeing results by chance
- Significance level: Typically $p < 0.05$
- Effect size: Practical importance of difference
- Power: Ability to detect actual differences
- Type I/II errors: False positives/negatives

ANOVA Analysis

Analysis of Variance: Testing differences across multiple groups

Understanding ANOVA

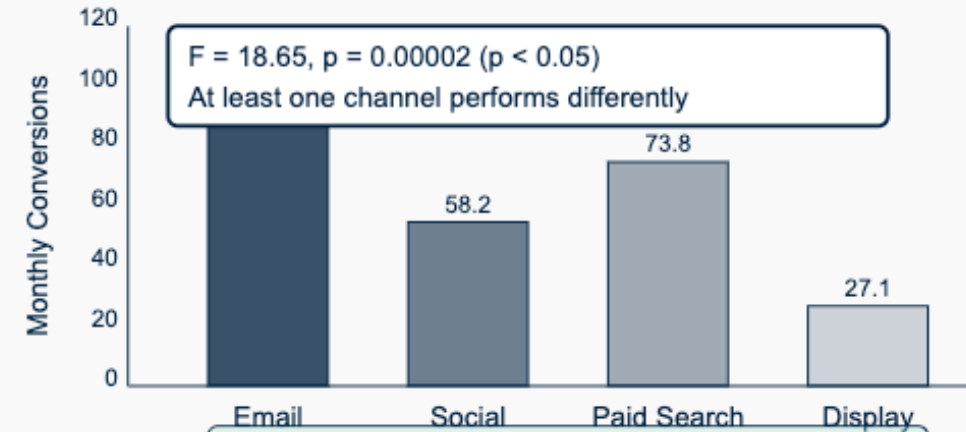
ANOVA (Analysis of Variance) examines:

- Variance BETWEEN groups
- Variance WITHIN groups



$$F\text{-statistic} = \text{Between Group Variance} / \text{Within Group Variance}$$

Business Case: Marketing Channel Performance



Post-hoc Tukey Test Results:

- Email vs. Display: Significant ($p < 0.001$)
- Email vs. Social: Significant ($p < 0.01$)
- Paid Search vs. Display: Significant ($p < 0.01$)
- Social vs. Paid Search: Not significant ($p = 0.08$)

When to Use ANOVA:

Technical Criteria

- Comparing three or more groups
- Continuous dependent variable
- Approximately normal distribution within groups
- Similar variance across groups (homoscedasticity)
- Independent observations

Types of ANOVA

- One-way ANOVA: One independent variable
- Two-way ANOVA: Two independent variables
- MANOVA: Multiple dependent variables
- Repeated measures ANOVA: Same subjects tested multiple times

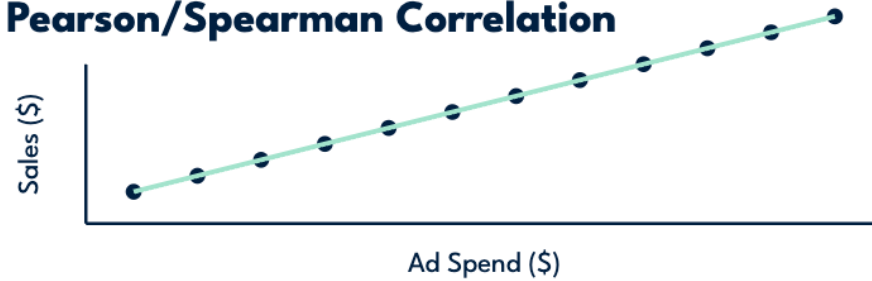
Business Applications

- Comparing multiple marketing channels
- Testing different product variations
- Analyzing regional performance differences
- Evaluating multiple pricing strategies
- Optimizing resource allocation decisions

C. Correlation & Relationship Analysis

Detect associations between metrics or behaviors

Pearson/Spearman Correlation



Example: Measuring the correlation between ad spend and sales ($r=0.78$), indicating a strong positive relationship

Scatterplots with Trendlines



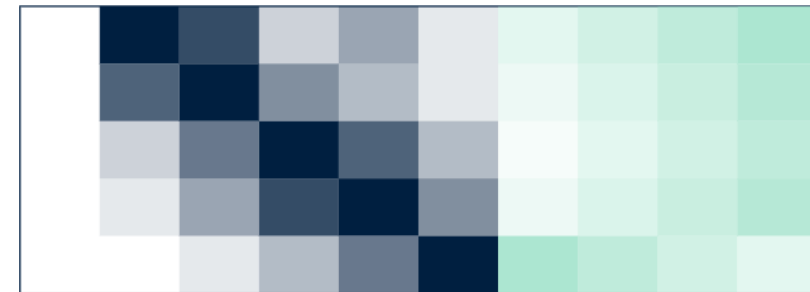
Example: Visualizing CAC vs LTV for different marketing channels to identify optimal investment points

Lagged Correlation Analysis



Example: Discovering that PR mentions correlate with sales increases 7-10 days later

Multivariate Correlation Heatmaps



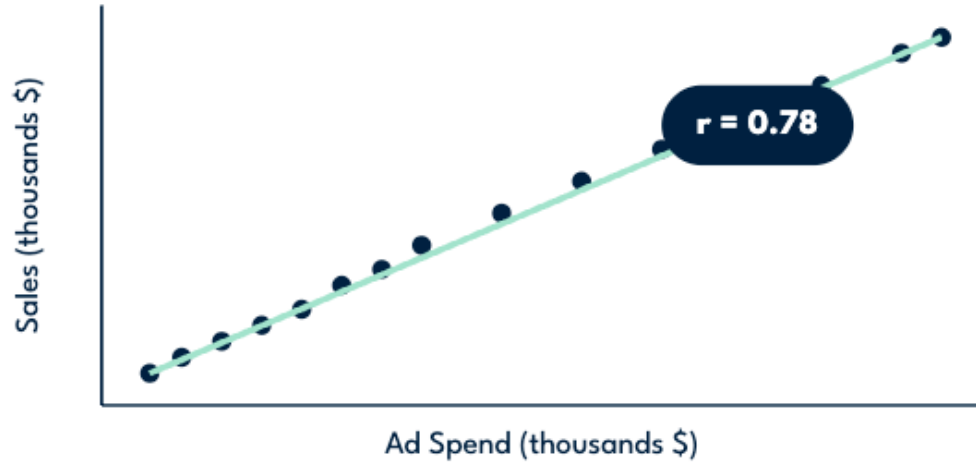
Example: Visualizing how different product features correlate with retention, revenue, and referrals

Business Value: Quantifies relationships between metrics to prioritize actions and understand key business drivers

Pearson/Spearman Correlation

Visualizing the strength and direction of relationships between metrics

Ad Spend vs. Sales Correlation



Understanding Correlation Coefficients

Key Components:

- Measures the strength and direction of linear relationships
- Values range from -1 (perfect negative) to +1 (perfect positive)
- The closer to -1 or +1, the stronger the relationship
- Pearson (r): measures linear relationship
- Spearman (ρ): measures monotonic relationship (rank order)



Business Applications

Marketing Optimization

- Identify most effective channels
- Optimize ad spend allocation
- Measure campaign impact
- ROI analysis

Product Development

- Feature impact analysis
- User satisfaction drivers
- Prioritize improvements
- A/B test result analysis

Customer Analysis

- Identify behavior patterns
- Segment customer base
- Predict future purchases
- Loyalty factor analysis

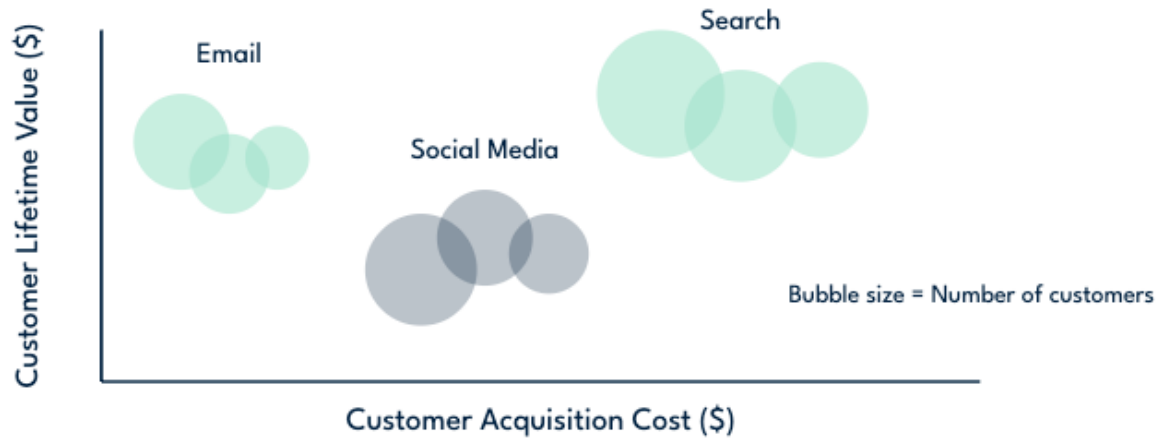
Implementation Tips

- Start with clear hypotheses before calculating correlations
- Check data for normality to choose Pearson vs. Spearman
- Remember correlation $\neq$ causation
- Look for spurious correlations due to third variables
- Visualize data alongside coefficient values
- Test statistical significance (p-value) of correlations

Scatterplots with Trendlines

Visualize relationships and identify optimal operating points

CAC vs. CLV by Marketing Channel



Understanding Scatterplots

Key Components:

- Each point represents an individual observation
- X and Y axes show two different variables
- Trendline shows overall relationship pattern
- Point size/color can add additional dimensions

Trendline Types:

- Linear: $y = mx + b$ (straight line)
- Polynomial: curved relationship
- Logarithmic: quick rise, then plateau
- Exponential: slow start, then rapid increase

Business Applications

Marketing Analysis

- Channel performance
- CAC vs LTV comparison
- Identify diminishing returns
- Optimize ad spend

Operations

- Price vs demand elasticity
- Production cost vs quality
- Inventory levels vs stockouts
- Logistics optimization

Financial Planning

- Revenue vs cost factors
- Investment performance
- Risk vs return analysis
- Budget allocation

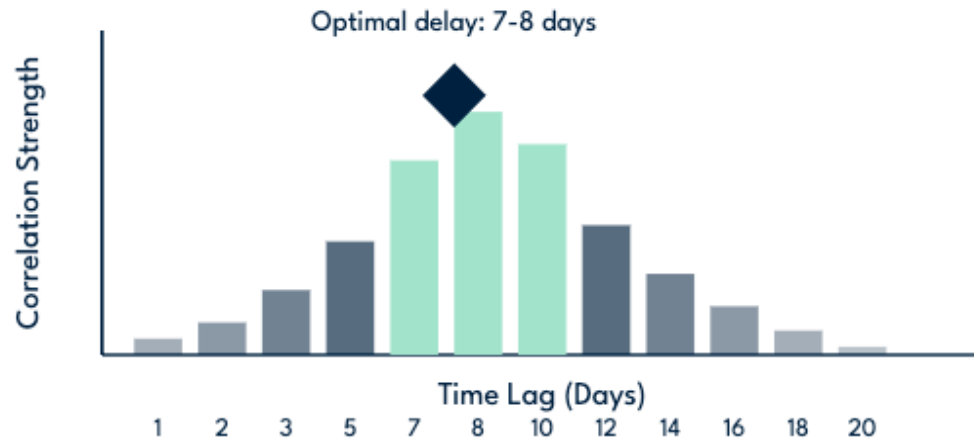
Implementation Tips

- Use color/size to add third and fourth dimensions
- Include R^2 value to show trendline fit quality
- Try different trendline types for best fit
- Label outliers for further investigation
- Use interactive tools to explore data points
- Consider segmenting data before plotting

Lagged Correlation Analysis

Account for time delays in cause-effect relationships

PR Mentions vs. Sales Increase



Understanding Lagged Correlation

Key Concept:

- Examines relationship between variables over time
- Tests if a change in X predicts a change in Y later
- Helps establish temporal ordering (which came first)
- Identifies optimal time delay between cause and effect

Formula:

- $\text{Corr}(X[t], Y[t+k])$ where k = time lag
- Test multiple k values to find strongest correlation
- Can use either Pearson or Spearman method
- Autocorrelation: comparing variable with itself over time

Business Applications

Marketing

- Ad spend → revenue impact
- PR mentions → brand lift
- Email campaigns → purchases
- Social activity → conversions

Sales & Operations

- Inventory levels → stockouts
- Price changes → demand shift
- Weather events → product sales
- Supply chain delays

Customer Behavior

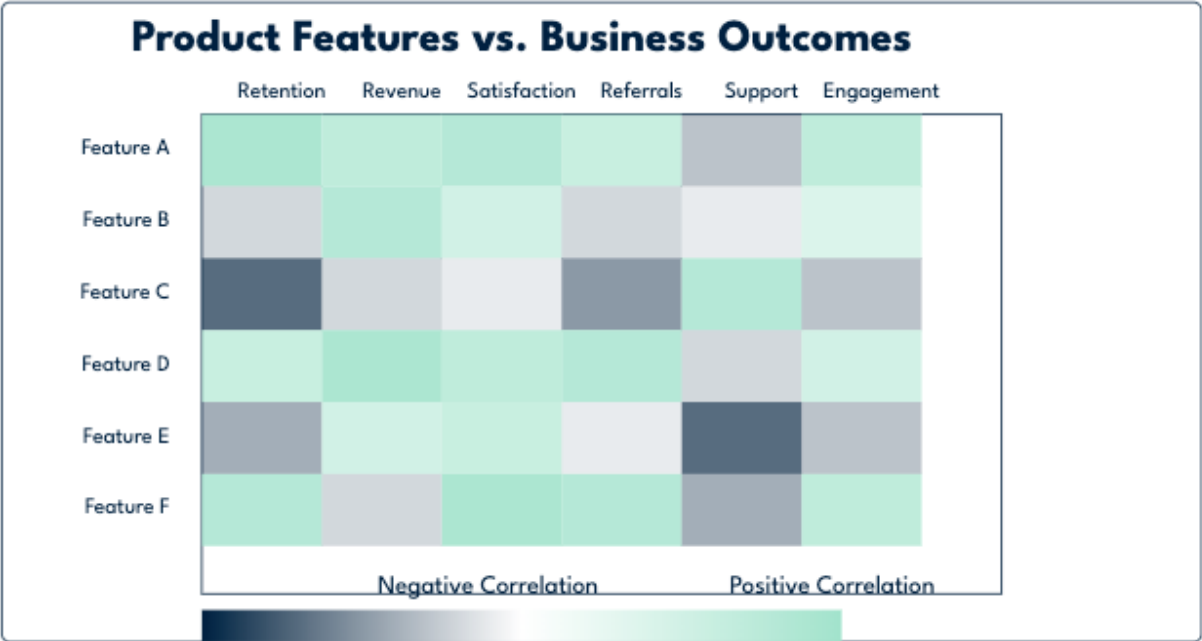
- Support issues → churn
- Feature usage → retention
- NPS changes → referrals
- Purchase patterns over time

Implementation Tips

- Test both small and large time lags
- Use consistent time intervals
- Control for seasonality and trends
- Consider industry-specific delay patterns
- Test statistical significance at each lag
- Visualize with correlograms or lag plots

Multivariate Correlation Heatmaps

Quickly spot strong relationships in complex datasets



Understanding Correlation Heatmaps

Key Components:

- Visual representation of correlation matrix
- Each cell shows relationship between two variables
- Color intensity represents correlation strength
- Different colors for positive vs. negative correlations

Advantages:

- Shows all variable relationships at once
- Quickly identifies strongest relationships
- Handles large number of variables efficiently
- Works with different correlation methods

Business Applications

Product Development

- Feature impact assessment
- Identify key satisfaction drivers
- Feature interaction effects
- Feature prioritization

Financial Analysis

- Stock portfolio optimization
- Risk factor correlation
- Economic indicator analysis
- Asset class relationships

Marketing Mix

- Channel effectiveness
- Campaign interaction effects
- Customer segment response
- Cross-channel synergies

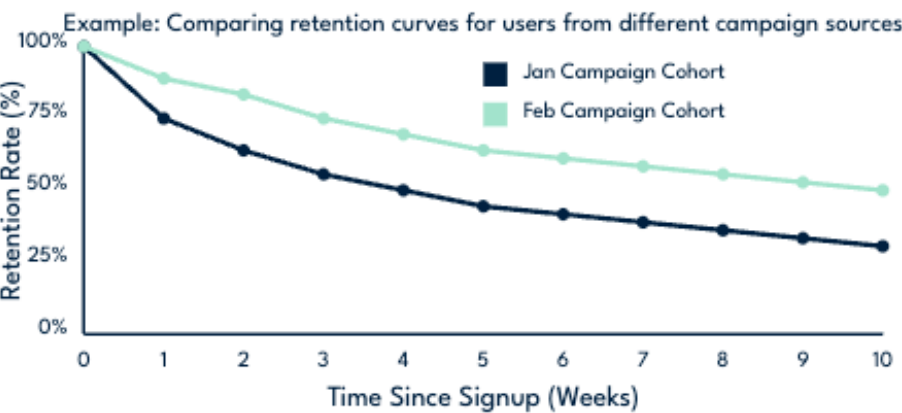
Implementation Tips

- Use clustered heatmaps for large datasets
- Include correlation values in cells for precision
- Sort variables by similarity to reveal patterns
- Use interactive tools for drill-down analysis
- Mark statistically significant correlations
- Focus on strongest correlations for actionable insights

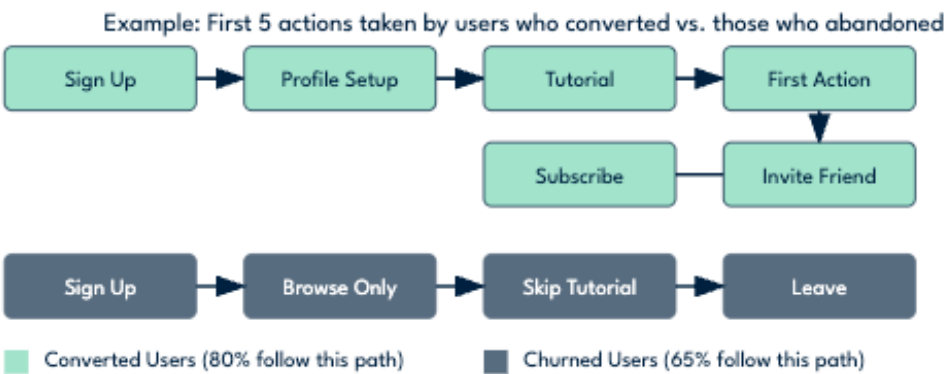
D. Cohort Behavior Deep-Dive

Understand how user behavior changes over time

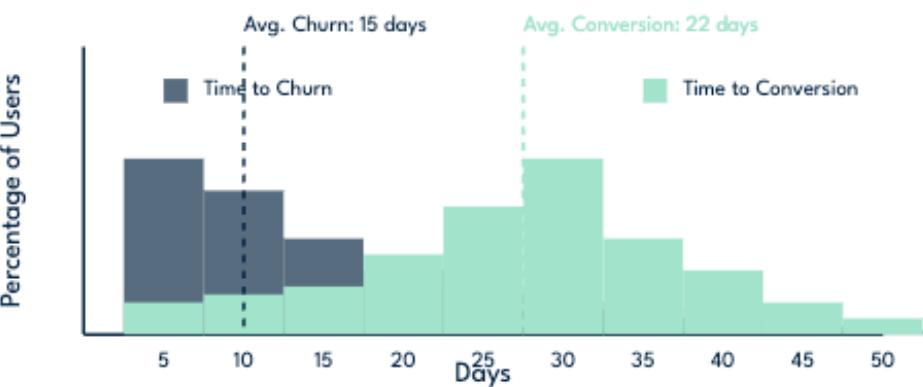
Behavioral Retention Comparison



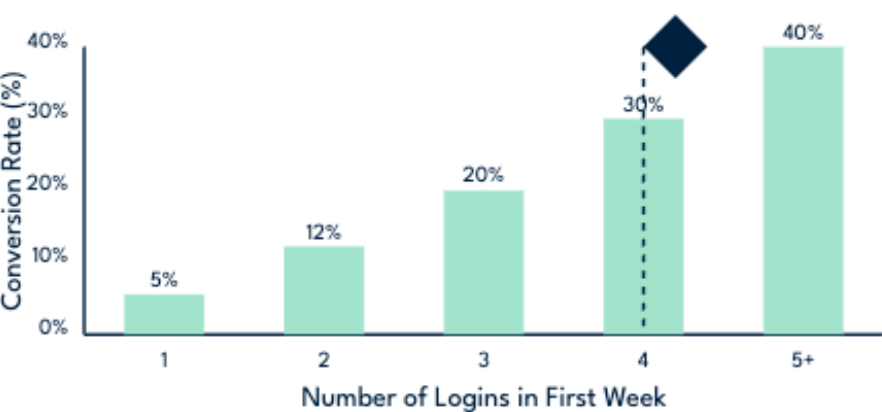
Event Sequence Pattern Comparison



Time to Event Analysis



Action Frequency Analysis

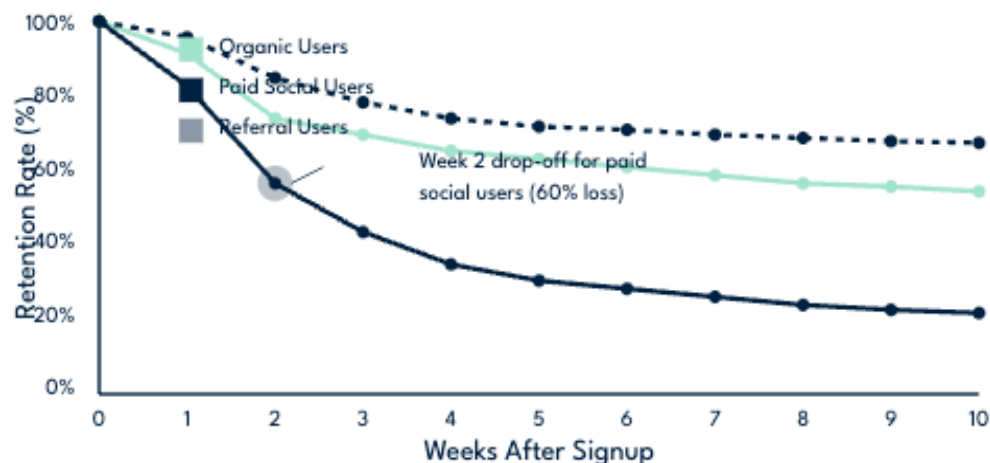


Business Value: Isolates acquisition quality and identifies critical behavior thresholds that predict outcomes

Behavioral Retention Comparison

Comparing retention curves for users who joined during different campaigns

User Retention by Acquisition Source



Understanding Retention Cohorts

Key Concept:

- Compares retention over time for different user groups
- Helps isolate acquisition quality from product issues
- Reveals effectiveness of acquisition channels
- Shows impact of onboarding changes on retention

Cohort Types:

- Acquisition source (organic, paid, referral)
- Acquisition time period (monthly, quarterly)
- Feature adoption (used vs. didn't use feature X)
- User segment (plan type, demographics)

Business Applications

Marketing Optimization

- Evaluate channel ROI
- Identify low quality traffic
- Target high-retention segments
- Optimize CAC with LTV data

Product Development

- Measure feature impact
- Test onboarding changes
- Identify drop-off patterns
- Improve value discovery

Business Strategy

- Long-term forecasting
- Evaluate business health
- Market expansion strategy
- Retention-based budgeting

Implementation Tips

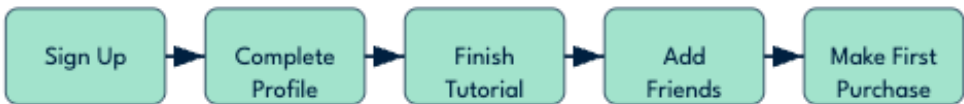
- Use consistent time intervals (weekly, monthly)
- Track both classic and rolling retention
- Control for seasonality when comparing cohorts
- Look for both initial drop-off and long-term slope
- Define meaningful retention events (not just logins)
- Target specific drop-off points with retention initiatives

Event Sequence Pattern Comparison

Analyzing the first actions taken by users who convert versus those who abandon

First 5 Actions Taken by Users

Converting Users (85% follow this path)



Churned Users (70% follow this path)



Key Finding: Profile completion is the most critical step!

Understanding Event Sequences

Key Concept:

- Compares the order and timing of user actions
- Identifies behavioral paths that lead to success or failure
- Reveals critical steps where users diverge
- Shows impact of skipping intended actions

Analysis Methods:

- Funnel flow analysis
- Markov chain modeling
- Process mining
- Sequence clustering

Business Applications

User Experience

- Optimize onboarding flows
- Redesign critical interfaces
- Streamline checkout process
- Improve success paths

Conversion Optimization

- Remove friction points
- Target intervention nudges
- Create guided product tours
- Automate follow-up triggers

Risk Management

- Detect fraud patterns
- Identify churn indicators
- Monitor compliance flows
- Predict user abandonment

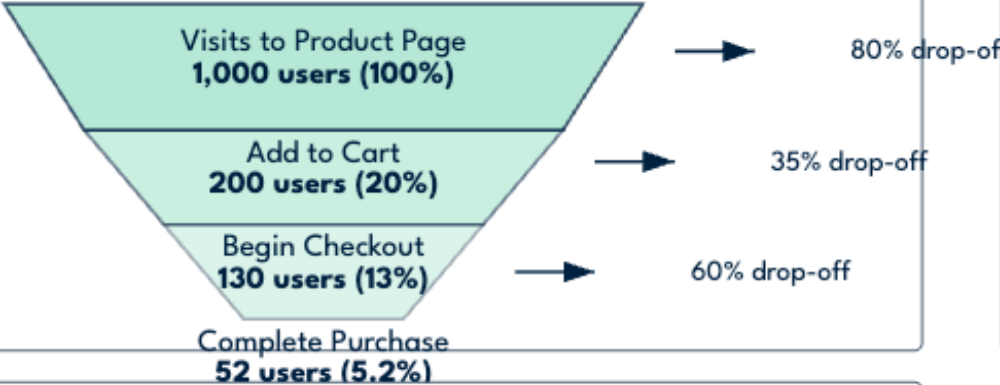
Implementation Tips

- Track both order and timing of events
- Focus on first-time experiences initially
- Segment by user type for meaningful comparisons
- Use heatmaps to visualize complex patterns
- Compare before/after product changes
- Test hypothesis by encouraging specific sequences

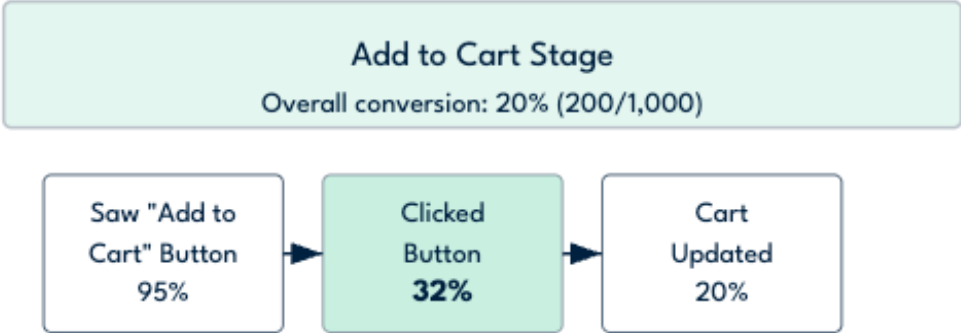
E. Funnel Drop-off Diagnosis

Pinpoint where users/customers are dropping off

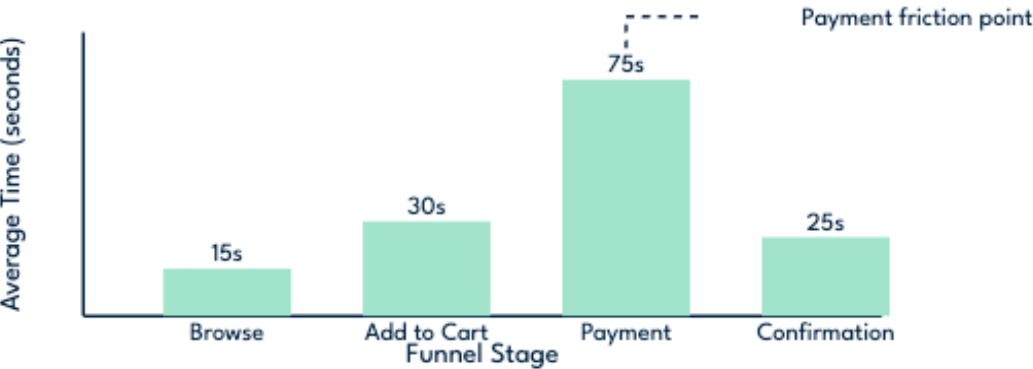
Stage-to-Stage Drop-off Rates



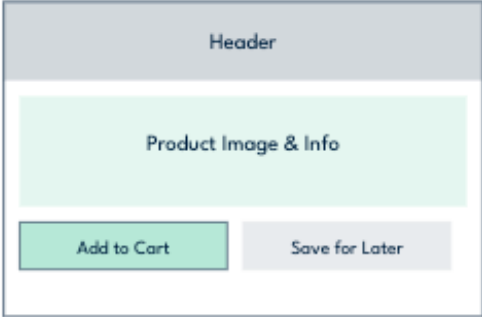
Micro-Conversion Analysis



Time Spent at Each Step



Clickmap/Scrollmap Interpretation



Key Findings:

- 85% of visitors see the Add to Cart button
- 32% of visitors click Add to Cart
- Product image receives good engagement
- "Save for Later" button gets minimal clicks
- 15% of users never scroll to see pricing

Business Value: Identifies the most problematic stage in conversion funnels to focus improvement efforts

Micro-Conversion Analysis

Breaking down a single funnel step into smaller measurable actions

Checkout Process Micro-Conversion Flow

Checkout Step: 52% Overall Conversion (130 → 68 users)



Key Findings:

- Contact info and shipping forms have good completion rates (92%, 84%)
- Payment form has the highest abandonment rate (32% drop)

Understanding Micro-Conversions

Key Concept:

- Breaks down major funnel steps into smaller measurable actions
- Identifies specific friction points within a single step
- Reveals where users encounter difficulties or hesitate
- Provides focused targets for optimization

Common Micro-Conversion Types:

- Form field completion (started vs. completed)
- Button visibility vs. clicks (saw button vs. clicked it)
- Sequential page views (viewed page 1 → viewed page 2)
- Element interactions (hover, click, scroll)

Business Applications

Form Optimization

- Reduce form fields
- Add clear field validation
- Fix error message clarity
- Create multi-step forms

Page Element Design

- Improve CTA visibility
- Optimize button placement
- Enhance UI feedback
- Use progress indicators

Content Strategy

- Clarify instructions
- Add trust indicators
- Address key objections
- Show expected outcomes

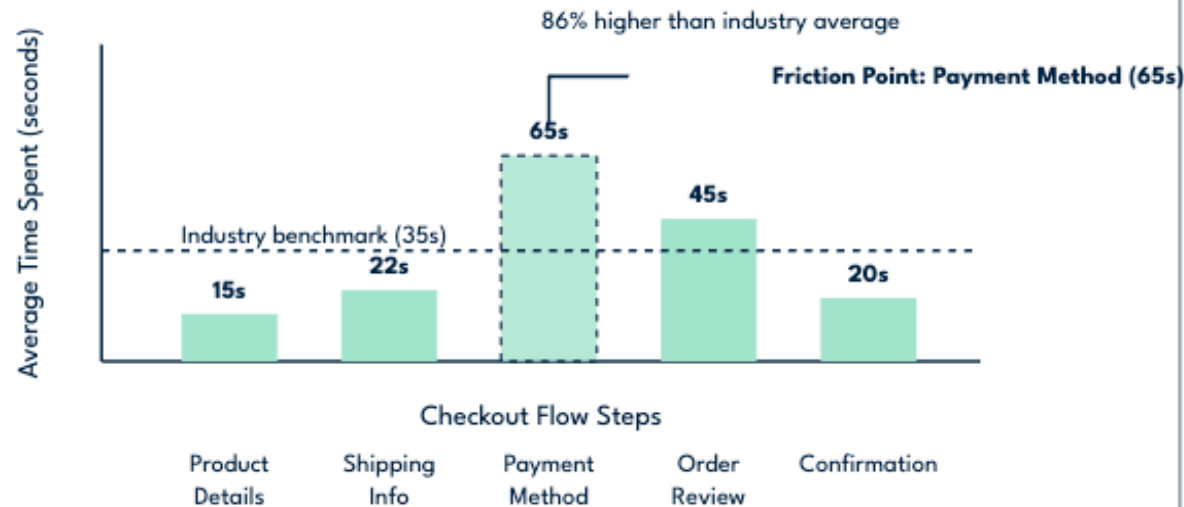
Implementation Tips

- Use analytics tool with event tracking capabilities
- Implement session recording for qualitative insights
- Define clear success metrics for each micro-step
- Test individual component changes in isolation
- Use form analytics to track field completion rates
- Set up alerts for unusual drops in micro-conversions

Time Spent at Each Step

Identifying friction points where users struggle to complete actions

E-commerce Checkout Flow: Time Spent Analysis



Understanding Time Spent Analysis

Key Concept:

- Measures how long users spend on each step of a process
- Identifies steps where users struggle or hesitate
- Highlights potential confusion or usability issues
- Provides context to drop-off rates

What It Reveals:

- Too Fast: Users may be skipping important information
- Too Slow: Potential confusion, complexity, or technical issues
- Expected Range: Process is likely functioning as designed
- Benchmark Comparison: Performance relative to standards

Business Applications

UX Improvements

- Streamline form elements
- Clarify field instructions
- Simplify complex choices
- Improve error handling

Technical Optimization

- Reduce page load times
- Optimize API calls
- Fix input validation delays
- Address browser issues

Process Redesign

- Reorder checkout steps
- Pre-fill known information
- Add express options
- Implement guest checkout

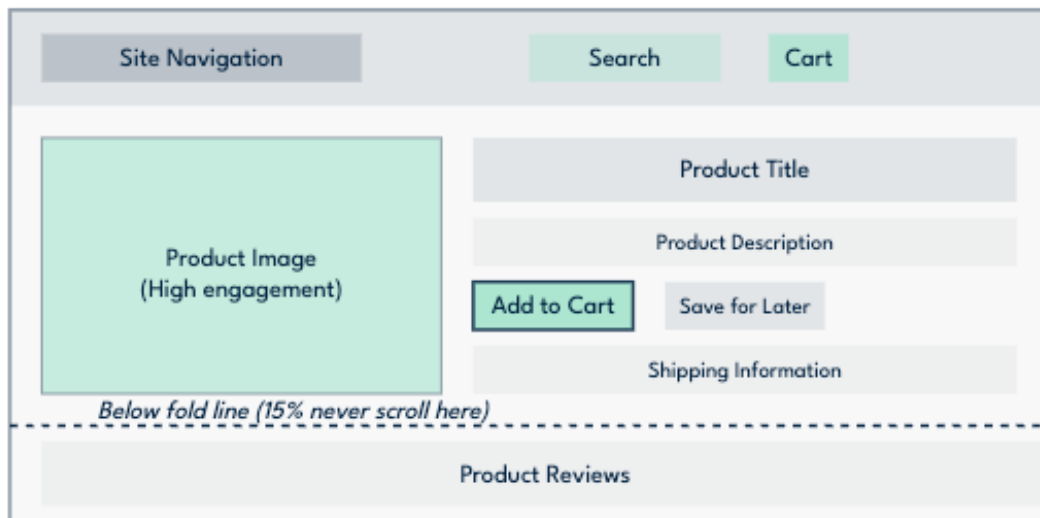
Implementation Tips

- Measure both average time and time distribution (percentiles)
- Segment analysis by device, browser, and user type
- Combine with session recordings to understand behavior
- Set appropriate time benchmarks for each step
- Compare successful vs. abandoned sessions
- Establish alerts for unusual time patterns

Clickmap/Scrollmap Interpretation

Revealing actual user interaction patterns vs. design assumptions

Product Page Clickmap Analysis



Key Findings from Example

- Primary CTA ('Add to Cart') receives strong engagement (32% click rate)
- 'Save for Later' button gets minimal interaction (3% click rate)
- Product image receives high attention (65% engagement)
- 15% of users never scroll below the fold to see reviews
- Cart icon in navigation has high visibility and click-through rate

Interpreting User Interaction Maps

Common Map Types:

- Clickmaps:** Show where users click or tap on a page
Reveals misplaced CTAs, distracting elements, missed opportunities
- Scrollmaps:** Visualize how far users scroll down a page
Identifies content visibility issues and engagement drop-off points
- Movemaps:** Track mouse movement patterns across a page
Shows attention patterns, reading flow, and hover behavior
- Attention maps:** Combine various signals to show overall attention
Provides comprehensive view of user engagement with content

Implementation Tips

- Use specialized heatmap tools (Hotjar, Crazy Egg, etc.)
- Collect sufficient sample size (1,000+ sessions minimum)
- Segment maps by traffic source, device type, and user status
- Compare before/after redesigns with the same methodology
- Analyze both successful and abandoned customer journeys
- Combine with session recordings for deeper qualitative insights

F. Experimentation & Incrementality Analysis

Find causality or uplift from interventions

A/B Test Diagnostic Metrics

Version A (Control)

Conversion: 5.2%
Sample: 10,000 users
Revenue/user: \$2.15

Version B (Variant)

Conversion: 5.3%
Sample: 9,800 users
Revenue/user: \$2.30

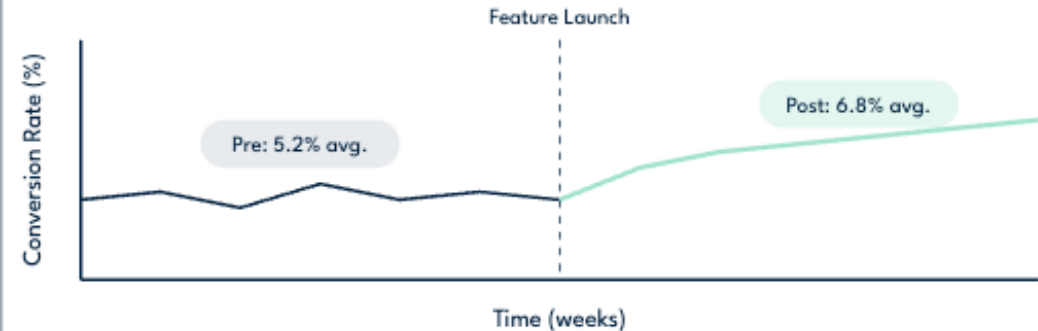
Overall: +1.9% conversion, +7.0% revenue (inconclusive)

Mobile: 4.1%

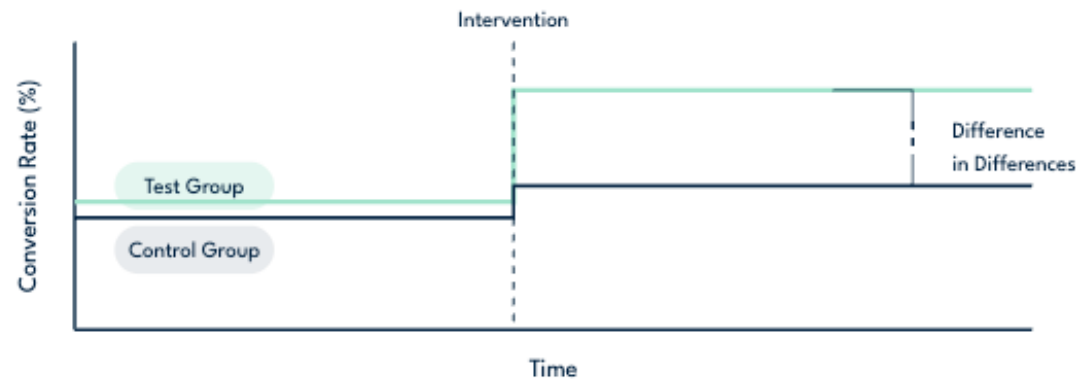
Mobile: 5.8% (+41%)

Significant improvement for mobile users!

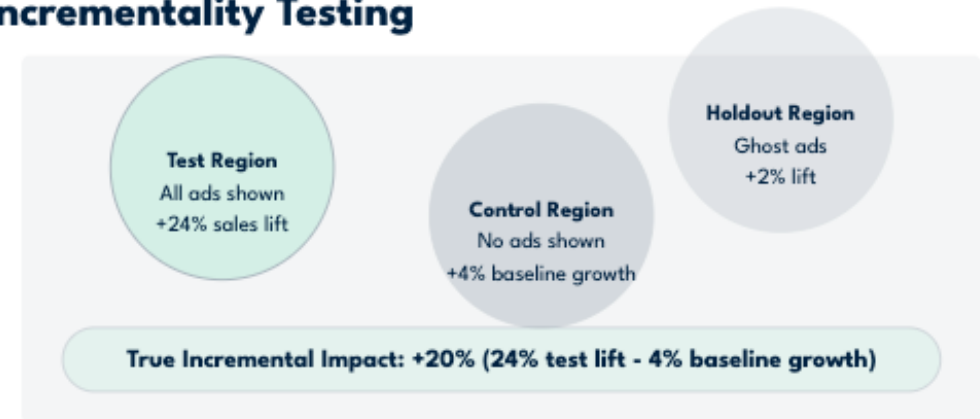
Pre-Post Analysis



Diff-in-Diff Analysis



Incrementality Testing



Business Value: Controls for external factors to get more accurate impact measurement

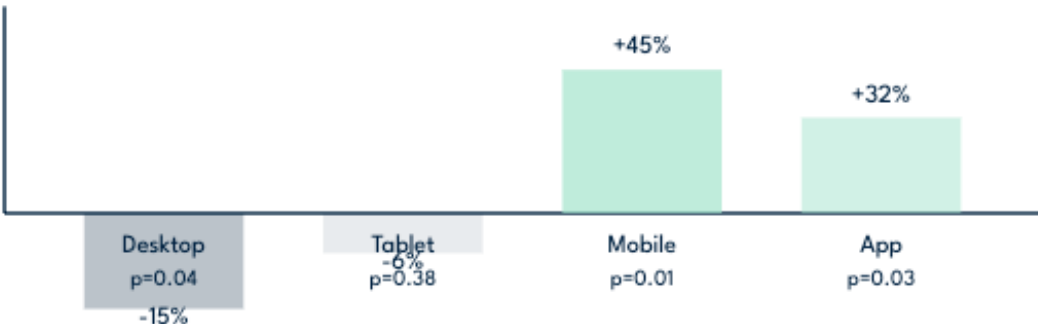
A/B Test Diagnostic Metrics

Breaking down test results to find hidden successes or segment-specific effects

Breaking Down "Failed" A/B Tests

Overall Results: Product Page Redesign

Control: 3.2% conversion | Variant: 3.3% conversion (+3.1%, $p=0.21$)



Understanding A/B Test Diagnostics

Key Concept:

- Examines A/B test results beyond just overall metrics
- Segments results by device, audience, behavior, etc.
- Identifies hidden successes in overall neutral results
- Reveals potential negative impacts on specific segments

Common Segmentation Dimensions:

- Device type (desktop, mobile, tablet, app)
- User type (new vs. returning, registered vs. guest)
- Traffic source (organic, paid, social, email)
- Demographics (age, location, language)

Business Applications

Test Optimization

- Salvage "failed" tests
- Target specific segments
- Detect harmful changes
- Refine test hypotheses

Product Strategy

- Create segment-specific UX
- Prioritize development areas
- Identify platform strengths
- Design adaptive interfaces

Revenue Optimization

- Deploy to high-value segments
- Maximize segment LTV
- Target high-potential users
- Create personalized journeys

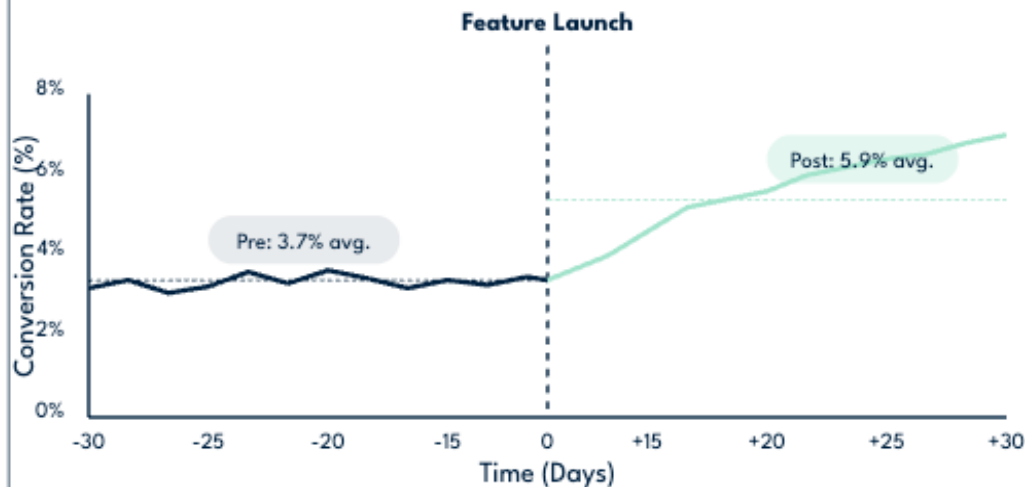
Implementation Tips

- Define segments before running the test (not after)
- Be cautious about multiple comparison problems
- Check segment sample sizes for statistical validity
- Look for both positive and negative segment impacts
- Validate findings with follow-up targeted tests
- Use consistent segment definitions across tests

Pre-Post Analysis

Comparing key metrics before and after a campaign or feature launch

Product Feature Launch Impact



Result: +59.5% increase in conversion rate (3.7% → 5.9%)

Understanding Pre-Post Analysis

Key Concept:

- Compares metrics before and after an intervention
- Useful when control groups aren't available
- Provides quick assessment of intervention impact
- Simple but vulnerable to external factors

Strengths & Limitations:

- ✓ Easy to implement
- ✓ Provides directional insight
- ✓ Effective for major changes
- ✓ Works with historical data

- ✗ Affected by seasonality
- ✗ Cannot control for trends
- ✗ External factors may bias results
- ✗ Correlation, not causation

Business Applications

Feature Launches

- Measure new feature impact
- Assess UX redesigns
- Evaluate pricing changes
- Track product improvements

Marketing Campaigns

- Assess campaign impact
- Measure brand initiatives
- Evaluate messaging changes
- Track conversion lift

Process Improvements

- Evaluate operational changes
- Measure support improvements
- Track efficiency gains
- Assess policy changes

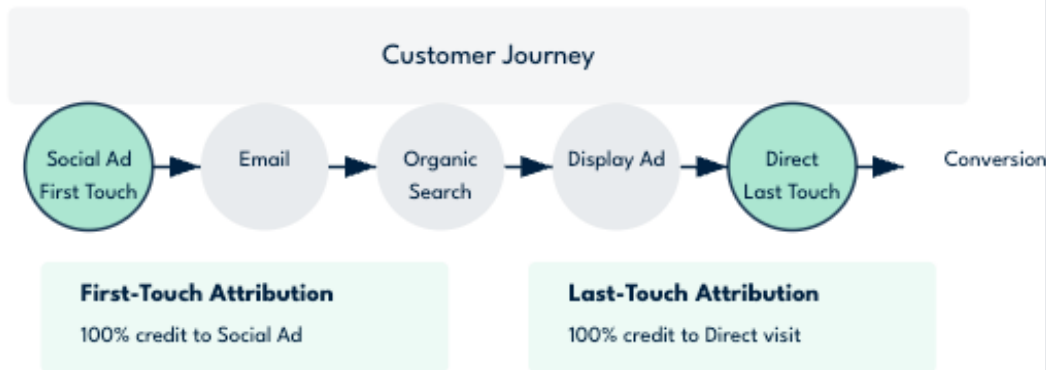
Implementation Tips

- Use equal time periods for pre and post (30/60/90 days)
- Control for seasonality by comparing to prior year
- Compare to overall industry trends when possible
- Look for step changes rather than gradual trends
- Segment analysis by user type or channel
- Use with caution for business-critical decisions

G. Attribution Analysis

Understand which inputs actually caused the result

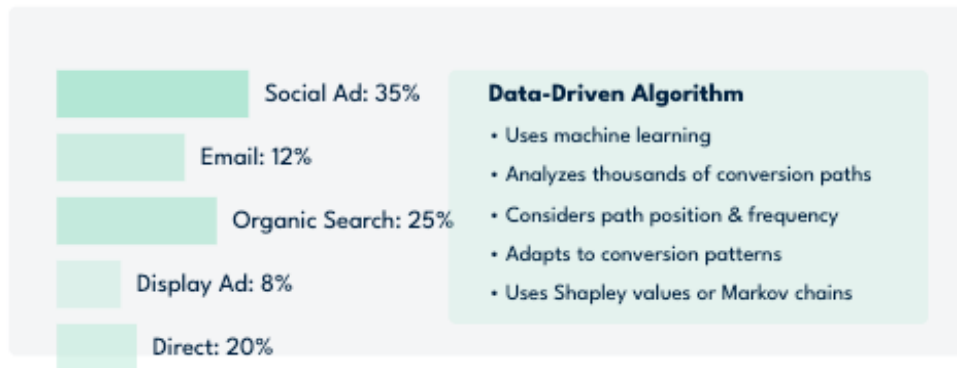
First/Last Touch Attribution



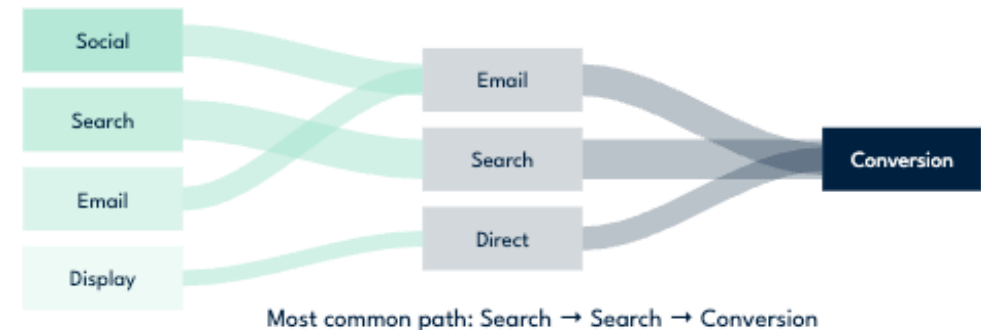
Multi-Touch Attribution



Data-Driven Attribution



Path-to-Conversion Analysis

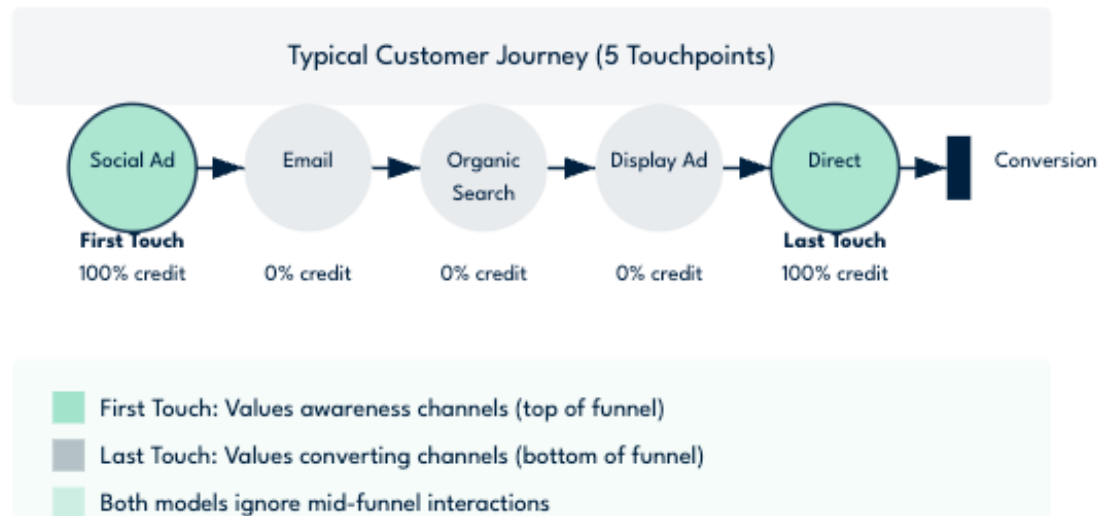


Business Value: Provides insights into channel effectiveness and conversion paths

First/Last Touch Attribution

Comparing attribution models that focus on entry and exit points

First vs. Last Touch Comparison



Understanding First/Last Touch

First Touch Attribution:

- 100% credit to initial interaction channel
- Best for understanding brand awareness
- Focuses on how customers discover you
- Simple to implement and understand

Last Touch Attribution:

- 100% credit to final interaction before conversion
- Best for understanding conversion drivers
- Focuses on closing channels
- Most commonly used default model

Business Applications

Marketing Strategy

- Assess channel effectiveness
- Top vs. bottom funnel spend
- Optimize media mix
- Quick ROI assessment

Budget Allocation

- Campaign planning
- Channel investment decisions
- Directional budget shifts
- Easy performance metrics

Channel Optimization

- Entry point improvement
- Closing channel efficiency
- Funnel coverage analysis
- Campaign performance

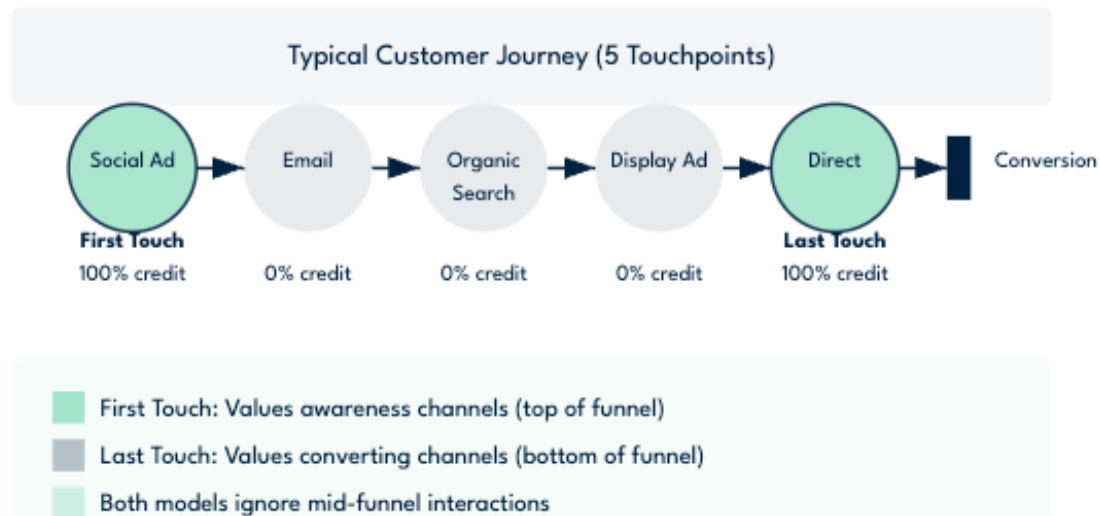
Implementation Tips

- Use both models in parallel to understand full funnel
- Set appropriate lookback windows (e.g., 30-90 days)
- Consider direct traffic classification issues
- Be aware of cross-device tracking limitations
- Use tracking parameters to improve attribution
- Compare with multi-touch models for complete view

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Financial Risk Management Case Study

Identifying & Assessing Risks with Data Analytics

What is Risk Management?

Understanding the fundamentals of identifying, assessing, and mitigating risk across business activities

Definition:

Risk management is the process of identifying, assessing, mitigating, and monitoring risks that could negatively impact an organization. Rather than eliminating all risks, the goal is to make informed decisions which risks to accept, reduce, or avoid. This allows businesses to operate effectively while managing uncertainty.



Identify

What could go wrong?



Assess

How bad could it be?



Mitigate

How can we reduce the harm?



Monitor

Are we staying on track?



Example:

Think of it like driving a car: You can't remove all risks, but you can wear a seatbelt, follow traffic laws, stay alert — all strategies to manage risk while moving forward.

→ Seatbelt: Mitigation → Road sign: Monitoring → Steering wheel: Informed Decision

The Role of Risk Management in Banking & Fintech

How effective risk management supports trust, compliance, innovation, and growth across financial services.

▷ Risk management plays several critical roles in the banking and fintech industry:



Traditional Banks

(Established Financial Institutions)

ensures stability & trust

- Ensures financial stability by addressing credit defaults
- Protects customer assets while maintaining trust
- Enables regulatory compliance with requirements
- Supports strategic risk-adjusted decision-making



Fintech Companies

(Innovative Digital Services)

balances innovation & security

- Facilitates innovation while mitigating vulnerabilities
- Builds trust in new and disruptive services
- Enables partnerships with traditional institutions
- Supports scalable growth by preventing risk events



Data Analytics

(Context Matters)

makes insights actionable

- Predictive modeling to anticipate potential risks
- ML algorithms to detect anomalies and patterns
- Real-time monitoring systems to identify risks
- Scenario analysis for various risk events

turns insights into effective risk management actions.

Example: A digital lending platform might use ML models to analyze thousands of data points for more accurate credit risk assessment.

The Six Pillars of Risk Management

Understanding the key risk types in the financial sector

▷ Financial risk management is typically categorized into six main pillars



The Six Pillars of Risk Management

Credit Risk

► Deep Dive into Credit Risk: Managing the Risk of Borrower Default in a Digital Lending Era

Definition

Credit risk is the potential for loss resulting from a borrower's failure to repay a loan or meet obligations.

Why It Matters

- Most significant risk for banks & fintechs
- Poor management → financial loss or institutional failure

Business Scenarios

- Digital lending to customers with limited credit history
- BNPL approval logic
- Credit cards for younger demographics

Data Applications

- Alternative data (e.g., utilities, rent, social media)
- Behavioral scoring
- Portfolio stress testing

Practical Solutions

- Real-time loan default trackers
- Advanced credit scoring using non-traditional data
- Risk-based customer segmentation

Common Challenges

- Balancing growth vs. default risk
- Lack of data for new-to-credit/underbanked customers
- Adapting models to economic shifts (e.g., COVID-19)

Real-World Example

BNPL firms like Afterpay and Klarna had to quickly adapt credit risk models during the COVID-19 pandemic—implementing shorter repayment terms, more frequent monitoring, and real-time behavioral indicators to manage rising credit risk.

The Six Pillars of Risk Management

Market Risk

► Zoom In: Market Risk — Navigating Volatility in a Digital Finance World

Definition

Market risk refers to the risk of losses due to fluctuations in interest rates, exchange rates, stock prices, or commodities.

Why It Matters

- Affects investments, trading, and loans with variable rates
- Volatility can strain liquidity or destabilize platforms

Business Scenarios

- Fintech stock trading platforms (e.g., Robinhood)
- Multi-currency neobanks
- Loan providers offering variable rates

Data Applications

- Value at Risk (VaR) for loss forecasting
- Sensitivity & correlation analysis
- Portfolio stress testing

Practical Solutions

- Real-time exposure dashboards
- FX volatility tracking
- Algorithmic hedging systems

Common Challenges

- Forecasting during black swan events
- FX risk in global operations
- Balancing growth vs. downside risk

Real-World Example

During the GameStop short squeeze in 2021, apps like Robinhood faced market risk at scale. Trading restrictions highlighted how market volatility also strains operational and liquidity risk management for fintech platforms.

The Six Pillars of Risk Management

Liquidity Risk

► Deep Dive into Liquidity Risk: Managing Cash Flow Stability in a Real-Time Financial World

Definition

Liquidity risk is the inability to meet short-term obligations due to cash shortage or inability to liquidate assets without loss.

Why It Matters

- Lack of liquidity → business failure despite profitability
- Impacts customer trust & regulatory compliance

Business Scenarios

- E-wallets handling user withdrawals
- P2P lending & merchant settlement
- Crypto yield platforms

Data Applications

- Predictive cash flow forecasting
- Liquidity stress testing
- Behavioral modeling of withdrawal patterns

Practical Solutions

- Real-time liquidity dashboards
- Dynamic buffer allocation
- Contingency funding frameworks

Common Challenges

- Growth > funding speed
- Withdrawal surges
- Cost of idle cash vs risk coverage

Real-World Example

BNPL firms like Afterpay and Klarna had to quickly adapt liquidity risk models during the COVID-19 pandemic—implementing shorter repayment terms, more frequent monitoring, and real-time behavioral indicators to manage rising cash flow demands.

The Six Pillars of Risk Management

Operational Risk

► Deep Dive into Operational Risk: Managing Process and System Reliability in Digital Finance

Definition

Operational risk involves potential losses resulting from inadequate or failed internal processes, people, systems, or external events.

Why It Matters

- As financial services become increasingly digital, operational resilience becomes critical
- System outages, process failures, or human errors can cause financial loss, reputational damage, and regulatory scrutiny

Business Scenarios

- A financial app experiencing system downtime during peak usage
- A digital bank protecting customer data from cybersecurity threats
- A payment processor ensuring transaction accuracy across millions of daily transactions

Data Applications

- Process Mining: Analyzing operational data to identify bottlenecks and failure points
- Anomaly Detection: Using machine learning to identify unusual patterns that might indicate operational issues
- Root Cause Analysis: Data-driven investigation of operational incidents to prevent recurrence

Practical Solutions

- Suspicious Activity Monitors: Systems that detect anomalies in operational processes or transactions
- Incident Management Platforms: Structured approaches to recording, resolving, and learning from operational incidents
- Business Continuity Planning: Comprehensive strategies for maintaining operations during disruptions

Common Challenges

- Keeping pace with rapidly evolving cybersecurity threats
- Maintaining system stability during periods of rapid growth
- Balancing innovation speed with operational risk considerations

Real-World Example

In April 2023, major UK digital bank Revolut experienced a significant operational incident when its app went down for several hours, preventing customers from accessing their accounts or making payments. The incident highlighted how operational risk has become central to digital banking, where a single technical issue can impact millions of customers simultaneously and trigger both reputational damage and regulatory attention.

The Six Pillars of Risk Management

Fraud/Behavioral Risk

▷ Deep Dive into Fraud/Behavioral Risk: Detecting Deceptive Activities in Digital Finance

Definition

Fraud risk encompasses potential losses from deceptive activities, while behavioral risk extends to customer behaviors that might create financial or reputational harm.

Why It Matters

- Digital financial services are prime targets for fraud, and the anonymous nature of online interactions makes detection challenging
- Understanding customer behavior patterns is essential for predicting risks and opportunities

Business Scenarios

- A payment app preventing account takeover and unauthorized transactions
- A digital bank detecting suspicious login attempts or unusual transaction patterns
- A lending platform identifying application fraud or synthetic identities

Data Applications

- Machine Learning Fraud Detection: Using AI to identify suspicious patterns across vast amounts of transaction data
- Behavioral Biometrics: Analyzing how users interact with devices (typing patterns, swipe behavior) to verify identity
- Network Analysis: Mapping connections between accounts to identify fraud rings

Practical Solutions

- Feedback Sentiment Trackers: Tools that analyze customer reviews and communications for signs of dissatisfaction or fraudulent intent
- Device Intelligence Systems: Solutions that analyze device characteristics and behaviors to identify risky devices
- Multi-layered Authentication: Risk-based approaches that apply appropriate security based on transaction risk

Common Challenges

- Balancing fraud prevention with customer experience (reducing friction)
- Adapting to rapidly evolving fraud techniques
- Managing false positives that can alienate legitimate customers

Real-World Example

During the COVID-19 pandemic, digital payment fraud increased dramatically as consumers shifted to online shopping. PayPal reported a 24% increase in daily transactions but also saw new fraud patterns emerge. The company enhanced its machine learning models to detect these new patterns, including account takeovers and fake selling schemes. They implemented additional security layers for high-risk transactions while maintaining a seamless experience for legitimate transactions, demonstrating the balance between fraud prevention and user experience.

The Six Pillars of Risk Management

Compliance/Regulatory Risk

► Deep Dive into Compliance/Regulatory Risk: Navigating the Complex Regulatory Landscape

Definition

Compliance risk refers to the potential for legal or regulatory sanctions, financial loss, or reputational damage due to failure to comply with laws, regulations, codes of conduct, or organizational standards.

Why It Matters

- The regulatory landscape for financial services is complex and constantly evolving
- Non-compliance can result in massive fines, business restrictions, and loss of trust

Business Scenarios

- A neobank ensuring KYC (Know Your Customer) compliance during digital onboarding
- A cross-border payments company navigating different regulatory requirements across jurisdictions
- A lending platform adhering to fair lending requirements and interest rate regulations

Data Applications

- Regulatory Reporting Analytics: Automated systems to ensure accurate and timely regulatory reporting
- Customer Risk Scoring: Data-driven approaches to determining customer risk levels for AML purposes
- Compliance Testing: Automated monitoring of transactions and activities to identify potential violations

Practical Solutions

- KYC Completeness Reports: Tools ensuring proper customer identification during digital onboarding
- AML Transaction Monitoring Systems: Platforms that flag suspicious activities based on defined scenarios and machine learning
- Regulatory Change Management Tools: Solutions that track regulatory developments and assess their impact on operations

Common Challenges

- Keeping up with rapidly evolving regulations across multiple jurisdictions
- Implementing compliance protocols without creating excessive customer friction
- Balancing innovation with regulatory constraints

Real-World Example

In 2020-2021, many fintech companies rushed to offer cryptocurrency trading and wallet services as crypto popularity surged. However, regulatory requirements around digital assets were (and still are) evolving rapidly. Companies like Coinbase invested heavily in compliance infrastructure, including enhanced KYC procedures, transaction monitoring systems specifically designed for crypto assets, and regular training for staff on evolving regulations. This proactive approach to compliance helped them navigate regulatory scrutiny better than many competitors.

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Credit Risk Management

Data Analysis Projects & Techniques

Practical approaches for real-world business problems

Key Credit Risk Analysis Projects

Top projects credit risk analysts undertake with business impact

Credit Scoring Models

Develop and refine statistical models to evaluate creditworthiness

Impact: 15-20% reduction in default rates



Portfolio Monitoring

Assess portfolio health and stress test under adverse scenarios

Impact: Proactive capital planning



Policy Optimization

Refine credit policies to balance risk and growth objectives

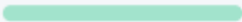
Impact: 10-15% profitability increase



Loss Forecasting

Estimate expected credit losses and calculate loan loss reserves

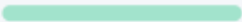
Impact: Efficient capital allocation



Alternative Data

Incorporate non-traditional data sources to enhance risk assessment

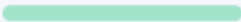
Impact: 20-30% market expansion



Fraud Detection

Identify potential fraud or early signs of credit deterioration

Impact: 40-50% fewer first-payment defaults



Regulatory Compliance

Ensure practices comply with regulations and prepare required reporting

Impact: Reduced regulatory penalties



Process Automation

Streamline credit risk processes while maintaining effective controls

Impact: 30-40% operational cost reduction



Strategy Segmentation

Develop different credit strategies for various customer segments

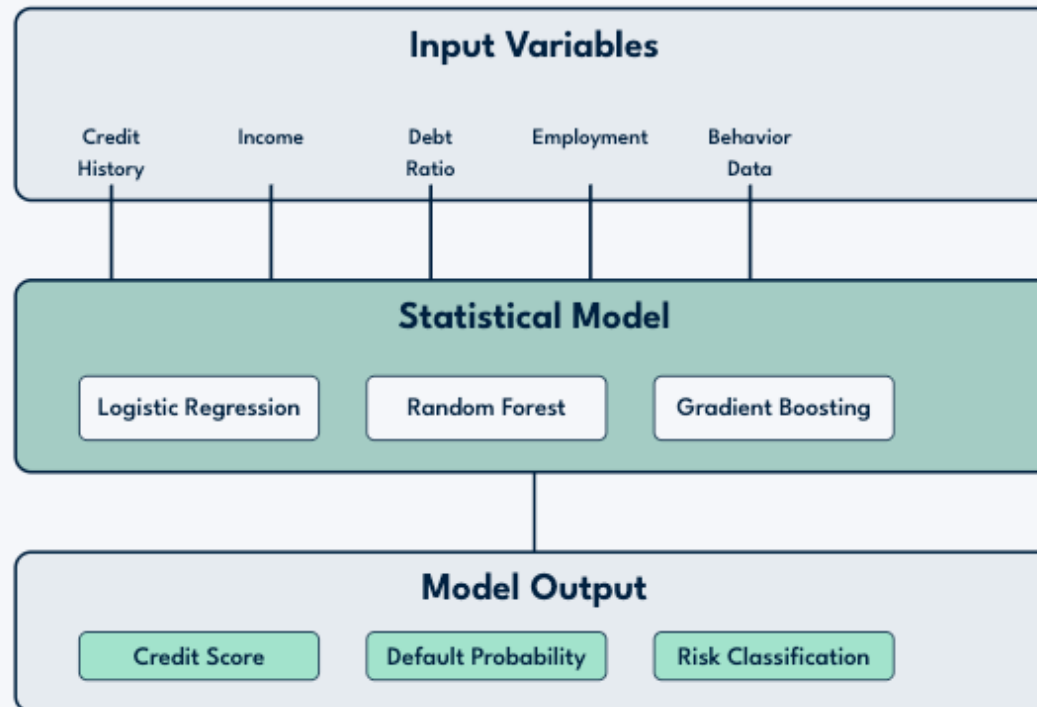
Impact: 20-25% higher approval rates



Credit Scoring Model Development

Developing and enhancing statistical models for evaluating creditworthiness

Credit Scoring Model Structure



Key Activities

- Selecting statistical methodologies
- Identifying predictor variables
- Building & testing models on historical data
- Implementing model governance
- Conducting regular performance reviews
- Developing specialized segment models

Business Impact

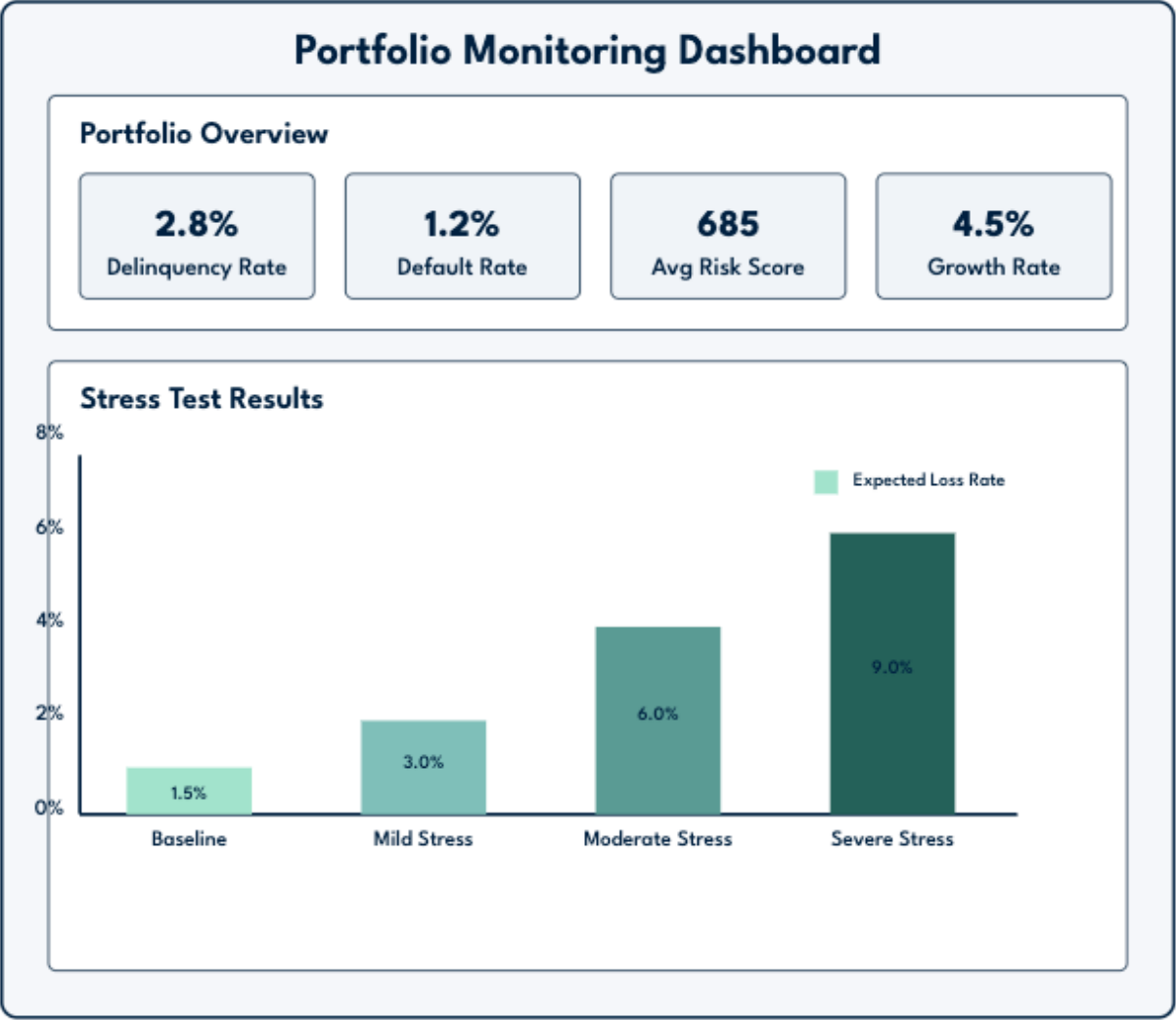
15-20% reduction in default rates
while maintaining or increasing approval rates

Key Deliverables:

- Statistical model documentation with feature importance
- Model performance reports (AUC, KS statistics)

Portfolio Monitoring & Stress Testing

Assessing portfolio health and resilience under adverse economic scenarios



Key Activities

Creating portfolio segmentation strategies

Developing early warning indicators

Designing stress testing methodologies

Modeling economic factor correlations

Creating scenario-based loss forecasts

Presenting results to management

Business Impact

Enables proactive portfolio management

and effective capital planning

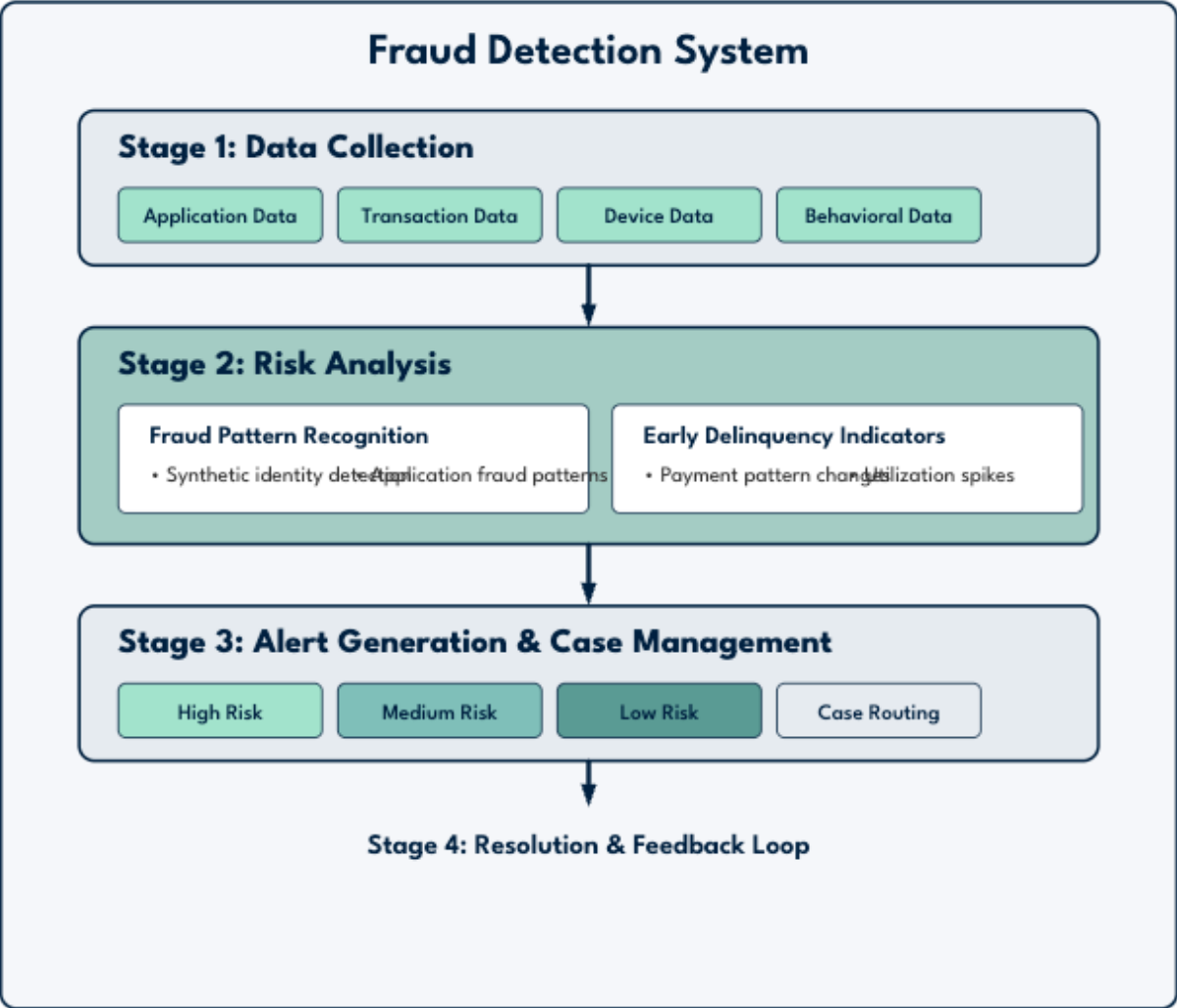
Key Deliverables:

• Monthly portfolio quality dashboards

• Stress testing results under various scenarios

Fraud & Early Delinquency Detection

Identifying potential fraud or early signs of credit deterioration



Key Activities

- Analyzing patterns of fraudulent applications
- Building models to detect synthetic identities
- Developing behavioral indicators of payment stress
- Creating alerting frameworks and workflows
- Measuring and optimizing detection effectiveness
- Implementing machine learning for detection

Business Impact

40-50% reduction in first-payment defaults

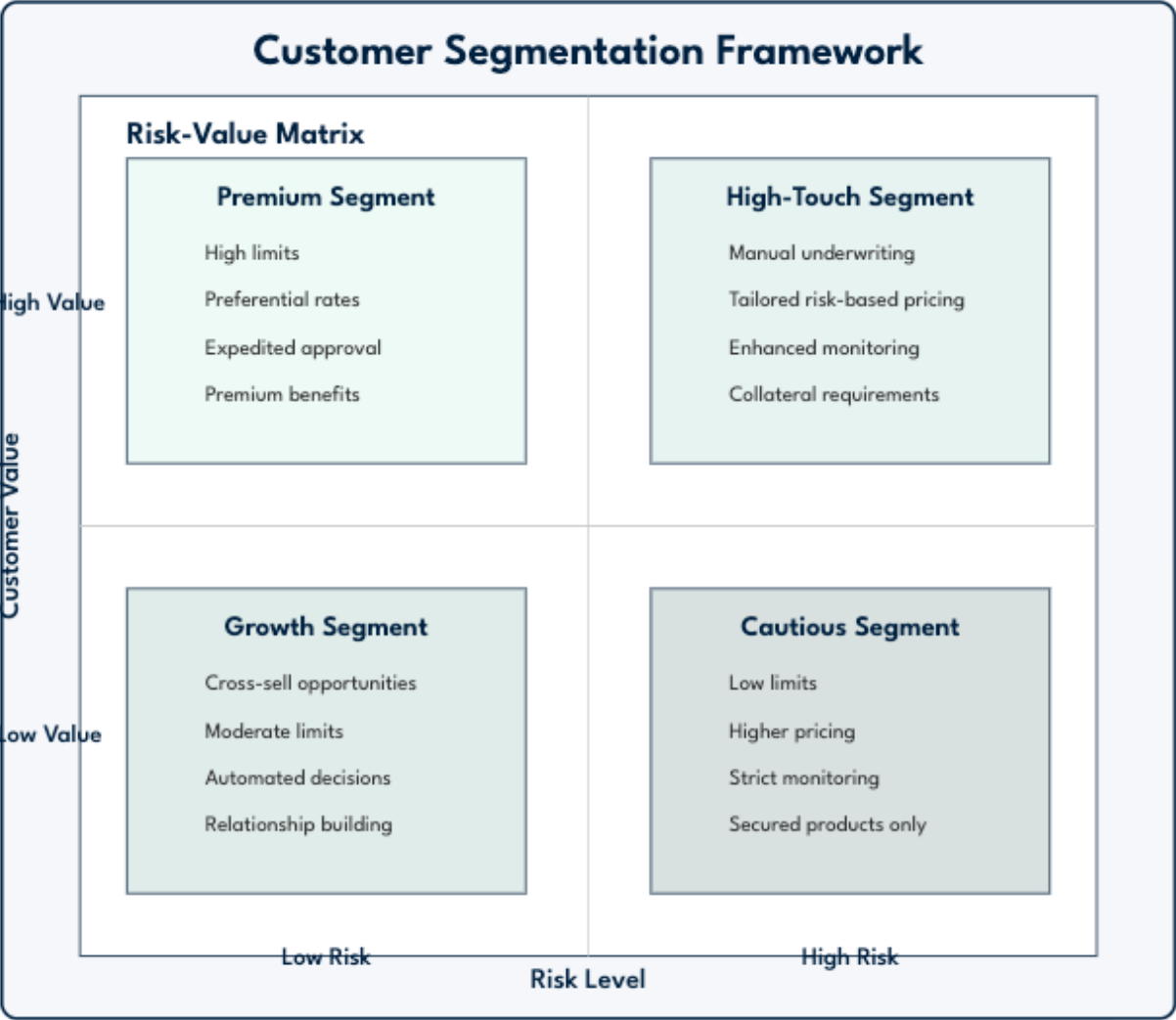
through effective early detection systems

Key Deliverables:

- Fraud pattern analysis reports
- Early warning system dashboards

Credit Strategy Segmentation

Developing differentiated credit strategies for various customer segments



Key Activities

- Creating customer segmentation based on risk
- Developing targeted acquisition strategies
- Building segment-specific underwriting
- Implementing test-and-learn frameworks
- Measuring segment performance
- Adjusting strategies based on performance

Business Impact

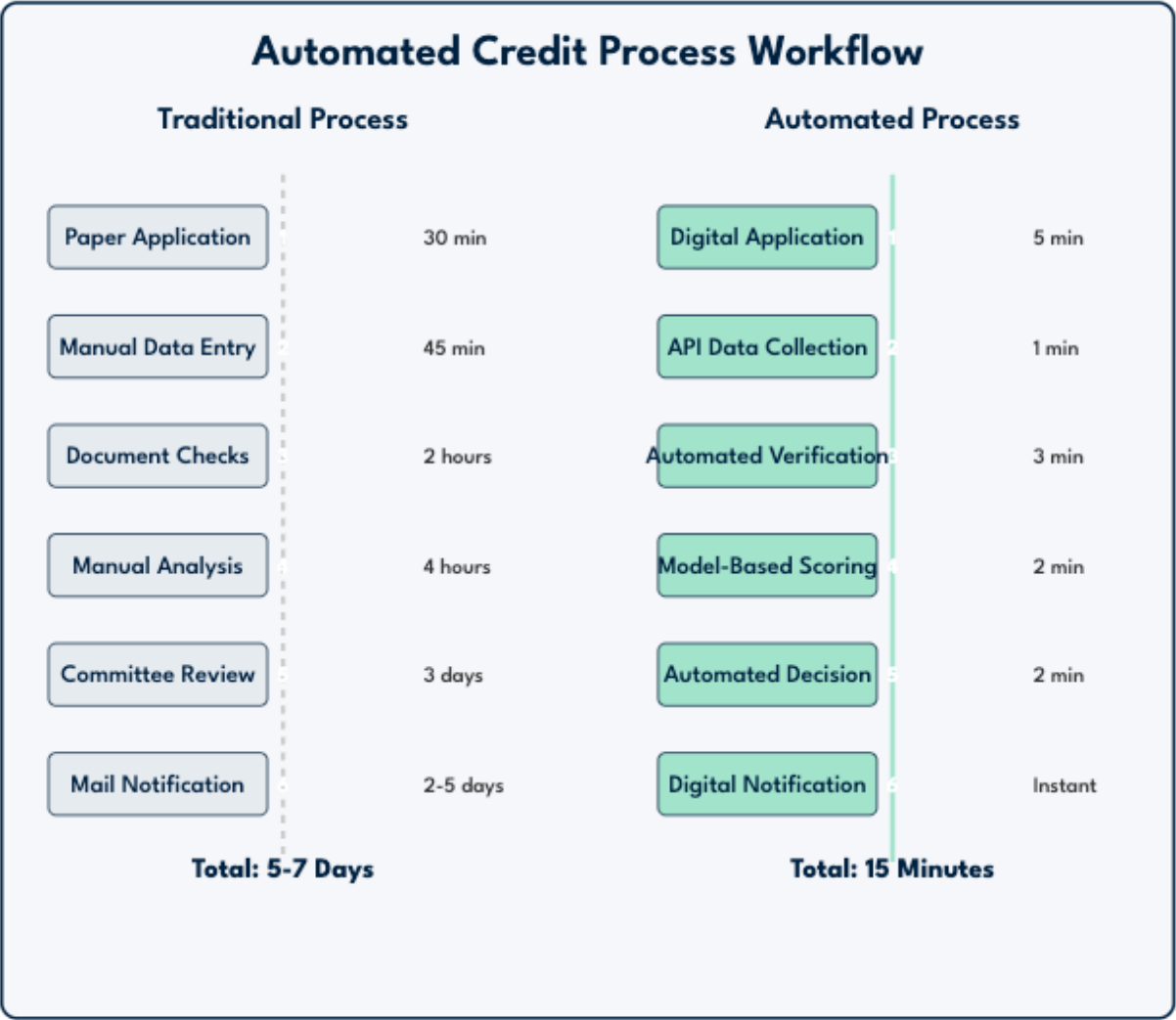
20-25% higher approval rates
through more granular segment strategies

Key Deliverables:

- Customer segmentation framework
- Segment performance dashboards

Credit Process Automation

Streamlining and automating credit risk processes while maintaining controls



Key Activities

- Mapping current-state credit processes
- Developing straight-through processing
- Implementing automated data collection
- Creating digital workflows for exceptions
- Establishing KPIs to measure efficiency
- Building integration with core systems

Business Impact

30-40% operational cost reduction

while improving customer experience

Key Deliverables:

- Process efficiency analysis
- Automation implementation roadmap

Credit Risk Analysis: Summary

Key projects and their business impact for credit risk analysts

Project	Key Activities	Business Impact
Credit Scoring Models	• Selecting statistical methodologies • Building & testing models on historical data • Conducting regular performance reviews	15-20% reduction in default rates
Portfolio Monitoring	• Creating portfolio segmentation strategies • Developing early warning indicators • Designing stress testing methodologies	Proactive capital planning
Fraud Detection	• Building models to detect synthetic identities • Developing behavioral indicators of payment stress • Creating alerting frameworks and workflows	40-50% less first-payment defaults
Strategy Segmentation	• Creating customer segmentation based on risk profiles • Developing targeted acquisition strategies • Implementing test-and-learn frameworks	20-25% higher approval rates
Process Automation	• Mapping current-state credit processes • Developing straight-through processing capabilities • Implementing automated data collection and verification	30-40% operational cost reduction

Risk Management Analysis Case

Data- Driven Risk Analysis: Leveraging historical loan data to optimize credit risk strategy

1 Business Context



- Peer-to-peer fintech platform connecting borrowers with investors from 2007-2014
- Revenue generated from borrower interest payments
- Dataset includes loan amounts, interest rates, credit grades
- Effective risk management directly impacts profitability

2 Project Goals



- Extract actionable insights from historical loan data
- Develop data-driven strategies to reduce credit risk
- Optimize interest rate structures based on risk categories
- Implement early warning systems for potential defaults
- Improve overall loan portfolio performance

3 EXPECTED OUTCOMES



Deeper Risk Understanding

Deeper understanding of borrower characteristics and risk factors

Helps build accurate profiles of high-risk customers

Identify common traits among defaulting borrowers



Data-Driven Credit Policy

Make lending decisions based on data models rather than intuition

Improve consistency and effectiveness in risk assessment

Support adjustments to approval criteria and loan limits



Better Risk Segmentation

More accurate categorization of borrowers according to risk levels

Increase accuracy when building risk prediction models

Customize approach for low, medium, and high-risk groups



Loan Portfolio Optimization

Improve overall performance of the loan portfolio

Direct impact on company revenue and profitability

Reduce default rates and increase debt recovery effectiveness

Understanding the Loan Dataset

Numerical Variables

- funded_amount
- investor_funds
- interest_rate
- annual_income
- debt_to_income_ratio
- delinquencies_2yrs
- public_record
- outstanding_principal

Categorical Variables

- term
- grade & sub_grade
- home_ownership
- income_status
- loan_status
- purpose
- state

Temporal Variables

- id & customer_id
- loan_issue_date
- earliest_credit_line

Derived Variables

- Loan age
- Credit history length
- Loan to income ratio
- Monthly payment
- Payment to income ratio